

bioMoR: Biology-Guided Mixture-of-Recursions for Effective Genomic Learning

Koushik Howlader¹, Tirtho Roy¹, Md Tauhidul Islam², Wei Le¹

¹Iowa State University, Iowa, USA

²Stanford University, CA, USA

Abstract

Transformer models for high-dimensional omics analysis process thousands of genes or pathways, although only a subset requires deep computation. Mixture-of-Recursions (MoR) improves efficiency through adaptive token-choice or expert-choice routing. We propose bioMoR, which, to the best of our knowledge, is the first framework to apply MoR to gene-level and pathway-level learning. Our contributions include identifying three locations for integrating structured biological knowledge within an MoR backbone: graph-based information sharing refines token embeddings, a structural bias guides self-attention toward biologically related tokens, and a graph-aware router uses neighborhood information to determine each token’s recursion depth. These techniques are centered on our insight that additional knowledge of token interaction can effectively help models construct embeddings and select which tokens should be learned more deeply. Across eight benchmarks spanning diverse omics data types and evaluated under a unified five-fold cross-validation protocol, bioMoR improves average macro-F1 by 8.2 percentage points and balanced accuracy by 7.1 percentage points over the strongest biology-agnostic MoR baseline while using 75 percent fewer parameters and up to 58 percent fewer FLOPs than a non-recursive Transformer. The selected marker genes or pathways provide biological interpretability, while their token-specific recursion depths reveal how computation is allocated.

1 Introduction

High-throughput sequencing now measures thousands of genes, pathways, or molecular features per sample. Transformers are attractive for these data because attention can model long-range molecular dependencies; recent single-cell foundation models, including scBERT (Yang et al. 2022), scGPT (Cui et al. 2024), scFoundation (Hao et al. 2024), and Geneformer (Theodoris et al. 2023), show strong results for annotation and representation learning. However, the standard Transformers spend the same deep computation on every token, even though genomic signal is usually sparse: a cell type, tumor state, or metastatic phenotype is often driven by a small set of marker genes, regulatory programs, or pathways. For computational efficiency and accuracy, a useful genomic model should not waste computation on weakly

informative features while under-refining the molecular programs that explain the phenotype. The effective modeling thus should answer two questions at the same time: which genes or pathways should be kept, and which of those tokens should receive more model computation?

Guiding assumption. We should perform computation and learning in a selective and structured manner, based on the semantics of omics data. The model should retain biologically meaningful marker gene/pathway tokens and spend more recursive computation on the tokens whose biological neighborhoods are most predictive for the task.

Existing methods address only part of this assumption. Efficient attention methods such as Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021), and Nyströmformer (Xiong et al. 2021) reduce sequence cost through low-rank or kernel approximations. Adaptive computation methods such as ACT (Graves 2016), mixture-of-experts (Shazeer et al. 2017), mixture-of-depths (Raposo et al. 2024), and mixture-of-recursions (Bae et al. 2025) route tokens dynamically. However, these approaches are biology-agnostic, for example, they do not know whether two genes express together, share a pathway, or contribute to a same biological process. Conversely, biology-aware resources and models such as CellMarker (Hu et al. 2023), PanglaoDB (Franzén, Gan, and Björkegren 2019), Reactome (Gillespie et al. 2022), genomic interaction graphs (Islam and Xing 2023), Pathformer (Liu et al. 2024), and scBiGNN (Yang et al. 2023) provide and consider useful semantic structures of omics data, however, they are usually used for feature engineering, architectural constraints, or post-hoc interpretation rather than for selecting tokens and deciding which tokens should receive deeper computation.

We propose **bioMoR**, a biology-guided Mixture-of-Recursions framework for genomic learning. To the best of our knowledge, it is the first work that integrates biology knowledge and Mixture-of-Recursion architecture to drive gene or pathway token selections for adaptive learning.

As shown in Figure 1, bioMoR first compresses high-dimensional inputs into marker-gene or Reactome pathway tokens. It then uses a shared recursive Transformer block and routes each token to an adaptive number of recursive steps. The key change from biology-agnostic MoR (Bae et al. 2025) is that we inject relevant biology knowledge into the model

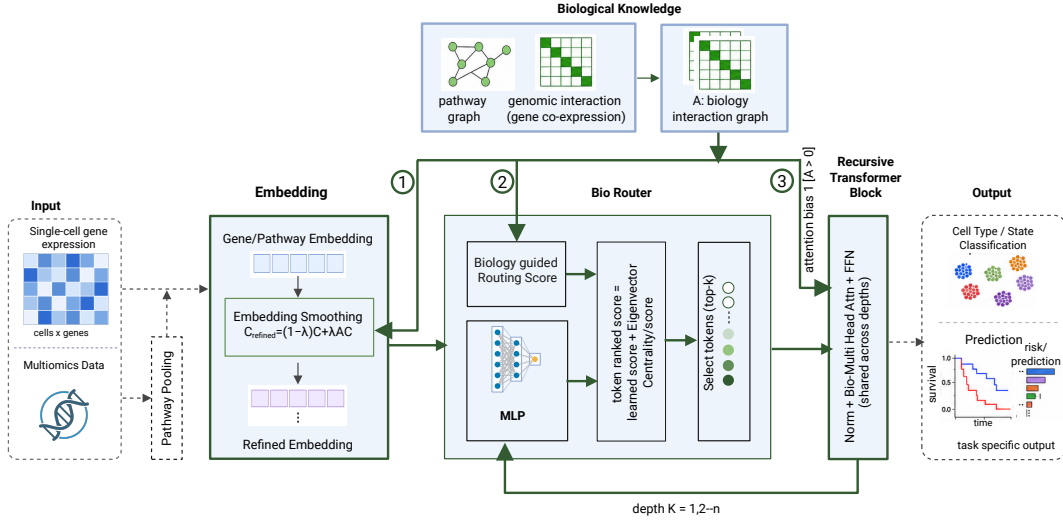


Figure 1: **Overview of bioMoR.** Single-cell and multi-omics inputs are represented as gene or pathway tokens. Fixed biological knowledge is incorporated at three sites: (1) embedding smoothing refines related tokens, (2) biology-guided routing selects tokens for further computation, and (3) attention bias promotes interactions between biologically related tokens in the weight-shared recursive Transformer block.

at three sites: it smooths token embeddings, biases attention toward related tokens, and guides the MoR router to decide token-specific recursion depth. Thus, biological structure and semantics becomes a control signal for adaptive computation, not only a static prior.

This leads to four questions in our evaluation: does bioMoR improve predictive performance (RQ1), which bio-knowledge injection site matters (RQ2), whether it indeed makes computation more efficient (RQ3), and which routing, marker budget, and model width settings are optimal (RQ4).

Across eight diverse genomic learning datasets, bioMoR improves average macro-F1 by 8.2 points and balanced accuracy by 7.1 points, while using 75% fewer parameters and up to 58% fewer FLOPs than traditional Transformers. It also achieved 0.92 linear-probe accuracy and successfully selected biologically relevant pathways for deeper computation, confirmed by prior literature.

In summary, our work makes the following contributions:

- We introduce bioMoR, the first Mixture-of-Recursions (MoR) framework for gene- and pathway-level omics learning.
- We integrated biology knowledge, including gene co-expression and pathway-pathway relations, to improve representation learning, attention learning, and adaptive routing leveraging MoR backbone.
- We conducted extensive evaluation and show that our techniques significantly improves F1 and balanced accuracy and reduce model parameters as well as the FLOP cost, and they are generally applicable across single-cell and multi-omics benchmarks.

2 Related Work

Genomic representation learning. Transformer models such as scBERT (Yang et al. 2022), scGPT (Cui et al. 2024), and Geneformer (Theodoris et al. 2023) learn gene representations from large single-cell datasets and support tasks

such as cell-type annotation. These models are effective, but normally apply the same model depth to every input token.

Biological knowledge in omics models. P-NET uses known biological hierarchies for cancer prediction (Elmarakeby et al. 2021), while Pathformer (Liu et al. 2024) and scBiGNN (Yang et al. 2023) use pathway or gene-interaction information to guide genomic learning. These methods show the value of biological knowledge, but they do not use it to decide how much computation each token should receive.

Adaptive computation. Adaptive Computation Time (Graves 2016), mixture-of-experts (Shazeer et al. 2017), mixture-of-depths (Raposo et al. 2024), and MoR (Bae et al. 2025) allow different tokens to receive different amounts of computation. However, their routing decisions are biology-agnostic. bioMoR connects these two research directions by using biological knowledge to guide token representations, attention, and recursion depth.

3 Method

First, in *Genomic Tokenization* and *Mixture-of-Recursions*, we provide background on two existing components (Cui et al. 2024; Elmarakeby et al. 2021; Bae et al. 2025), on top of which we built bioMoR. Second, in *bioMoR*, we explained our novel techniques of integrating biological knowledge to guide MoR learning. See Figure 1 for our overall workflow.

Genomic Tokenization

Let $x \in \mathbb{R}^{N \times C}$ denote one genomic sample with N genes and C molecular channels. Single-cell gene expression measures transcript abundance and has $C=1$ (Wang, Gerstein, and Snyder 2009). Multi-omics inputs may additionally contain mutation indicators, which record DNA-sequence alterations (Vogelstein et al. 2013), and copy-number variation (CNV), which records gains or losses of DNA segments (Feuk, Carson, and Scherer 2006). $T = [c_1, \dots, c_M] \in \mathbb{R}^{M \times d}$.

Single-cell marker tokens. A single cell has thousands of genes, but only a few are informative. We therefore turn the N genes into $M \ll N$ marker tokens. This makes the later attention cheap ($\mathcal{O}(M^2d)$) and lets each token be read as one named marker gene. First, each gene i is embedded as an identity vector plus its expression value,

$$t_i = e_i + W_v x_i, \quad e_i \in \mathbb{R}^d, W_v \in \mathbb{R}^{d \times C}, \quad (1)$$

following the gene-plus-value scheme of single-cell foundation models (Cui et al. 2024; Theodoris et al. 2023). We then keep M learnable queries; query m softly picks genes through keys $k_i = W_k e_i$,

$$w_{mi} = \frac{\exp(q_m^\top k_i / (\tau \sqrt{d}))}{\sum_{j=1}^N \exp(q_m^\top k_j / (\tau \sqrt{d}))}, \quad (2)$$

and sums them into a token $c_m = \sum_{i=1}^N w_{mi} t_i$. At test time, query m collapses to its top gene $g_m = \arg \max_i w_{mi}$, so every token names a marker gene.

Pathway tokens. For multi-omics cancer cohorts we instead group genes into Reactome pathways, which represent curated sets of genes participating in the same biological process (Gillespie et al. 2022). Each token is one pathway, built by pooling its member genes—mean pooling for dense expression and copy-number channels, and sum/burden pooling for sparse mutation channels. Every token is thus a named pathway.

Mixture-of-Recursions

Let $T = [c_1, \dots, c_M]$ be the marker/pathway token matrix. MoR applies one shared Transformer block f_θ up to K times:

$$H^{(0)} = T, \quad H^{(t+1)} = f_\theta(H^{(t)}), \quad t = 0, \dots, K-1. \quad (3)$$

Sharing makes parameter count independent of K , unlike K independent Transformer blocks (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025). A router decides which tokens continue. Expert-choice keeps the top- $\lceil c_t M \rceil$ tokens at step t ; token-choice lets each token self-select a depth with load balancing (Shazeer et al. 2017). The number of active steps,

$$d_m = \sum_{t=1}^K \mathbf{1}\{m \text{ is kept at step } t\}, \quad (4)$$

is the token-specific recursion depth.

bioMoR

Algorithm 1: bioMoR forward pass.

Input: x , biological knowledge B
token budget M , max depth K
 $T \leftarrow$ marker-gene or pathway tokens from x
 $A \leftarrow$ row-normalized fixed knowledge matrix from B
 $H \leftarrow (1 - \lambda)T + \lambda AT$; $\mathcal{A} \leftarrow \{1, \dots, M\}$
for $t = 1, \dots, K$ **while** $\mathcal{A} \neq \emptyset$ **do**
 $H_{\mathcal{A}} \leftarrow f_\theta(H_{\mathcal{A}}; A)$ *reuse the same block*
 $r_{\mathcal{A}} \leftarrow$ biology-aware router scores
 $\mathcal{A}_{\text{next}} \leftarrow$ tokens routed to the next step
tokens in $\mathcal{A} \setminus \mathcal{A}_{\text{next}}$ stop and keep current states
 $\mathcal{A} \leftarrow \mathcal{A}_{\text{next}}$ *return kept tokens to recursion*
end for
 $\hat{y} \leftarrow$ classifier(pool(H))
Train: minimize class-weighted CE plus auxiliary losses
Output: $\hat{y}$, markers/pathways, recursion depths d_m

Biological-knowledge construction. Following Genomap (Islam and Xing 2023), we construct a gene *co-expression* matrix that summarizes pairwise gene relationships. Its entries are partial correlations computed from the inverse covariance of the gene-expression matrix and restricted to the selected markers; this matrix forms the single-cell biological knowledge B . For multi-omics data, we use curated Reactome pathway relationships and treat them as undirected: $B_{ij} = B_{ji} = 1$ when pathways i and j are related, and both entries are zero otherwise (Gillespie et al. 2022).

Let D be the diagonal matrix of row sums of B . We use the fixed, row-normalized matrix

$$A = D^{-1}B. \quad (5)$$

The same biological knowledge is used for embedding smoothing (Site 1), attention bias (Site 2), and depth routing (Site 3).

Site 1: biology-guided embedding smoothing. Before recursion, we smooth the tokens using the biological-knowledge graph A , so that information of biologically related tokens is also integrated into the embedding of the token:

$$\tilde{T} = (1 - \lambda)T + \lambda AT, \quad (6)$$

where λ is learned. This denoises each marker/pathway token toward its biological neighbours before the repeated recursive updates.

Site 2: attention bias. Inside the recursive block, the *same* biological-knowledge graph A biases self-attention:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \lambda_{\text{attn}} \mathbf{1}[A > 0]\right) V. \quad (7)$$

This encourages biologically related tokens to exchange information during refinement.

Site 3: biology-aware routing. The original MoR router uses the current token representation $h_m^{(t)}$ to decide whether token m should continue to the next recursion step. bioMoR keeps this original score and adds a learned correction based

on the token’s biological neighborhood:

$$\hat{r}_m^{(t)} = \frac{w_r^\top h_m^{(t)}}{\tau_r} + \sigma(\gamma_t)\phi_\psi\left([AH^{(t)}, H^{(t)} - AH^{(t)}]_m\right). \quad (8)$$

The first term is the original MoR routing score. In the second term, $AH^{(t)}$ summarizes information from biologically related tokens, while $H^{(t)} - AH^{(t)}$ shows how the token differs from its biological neighborhood. The small network ϕ_ψ uses these two signals to adjust the routing score, and $\sigma(\gamma_t)$ controls how much correction is added at recursion step t . Because ϕ_ψ is initialized to output zero, the correction is zero at the start of training and bioMoR initially behaves like biology-agnostic MoR. During training, the correction becomes nonzero when biological information helps reduce the task loss; otherwise, it can remain close to zero. This design prevents noisy biological knowledge from affecting routing at the beginning while allowing the model to use it when helpful.

Training and efficiency. The main task loss is class-weighted cross-entropy. During training we minimize

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha\mathcal{L}_{\text{suf}} + \beta\mathcal{L}_{\text{div}} + \gamma\mathcal{L}_{\text{cmp}} + \eta\mathcal{L}_{\text{router}}, \quad (9)$$

where the auxiliary terms encourage marker sufficiency ($\mathcal{L}_{\text{suf}}$), marker diversity ($\mathcal{L}_{\text{div}}$), marker compression/sparsity ($\mathcal{L}_{\text{cmp}}$), and stable router logits and balanced token-choice routing ($\mathcal{L}_{\text{router}}$). In expert-choice routing, each recursion step selects a fixed number of the highest-scoring tokens. This fixed capacity directly controls how many tokens are processed at each step, so an additional load-balancing loss is not needed (Zhou et al. 2022; Bae et al. 2025). To quantify computational cost, we report FLOPs, a hardware-independent count of the arithmetic operations performed by the Transformer stack (Schwartz et al. 2020):

$$\Phi_{\text{eff}} = \sum_{t=1}^K (4a_t^2d + 4a_tdd_{\text{ff}}), \quad (10)$$

where a_t is the mean number of active tokens at recursion step t , d is the model width, and d_{ff} is the hidden dimension of the feed-forward network.

4 Experimental Setup

Datasets and tasks. We evaluated bioMoR on 8 datasets: Genomap single-cell data (Islam and Xing 2023), Reactome-based cancer data (Gillespie et al. 2022), and TCGA cohorts (The Cancer Genome Atlas Research Network et al. 2013) downloaded through Xena (Goldman et al. 2020). Single-cell tasks are multi-class cell-type classification; cancer tasks include binary classification and survival prediction on HNSC (head and neck squamous cell carcinoma), UCEC (uterine corpus endometrial carcinoma), COAD (colon adenocarcinoma), and KIRP (kidney renal papillary cell carcinoma) (The Cancer Genome Atlas Research Network et al. 2013), measured by Harrell’s C-index (Harrell et al. 1984). Gene expression measures RNA abundance (Wang, Gerstein, and Snyder 2009), mutation denotes a DNA-sequence alteration (Vogelstein et al. 2013), and copy-number variation (CNV) denotes gains or losses of DNA segments (Feuk, Carson, and

Scherer 2006). Single-cell inputs use gene tokens. For cancer, we create each pathway token by pooling the features of genes assigned to that Reactome pathway: mean pooling for gene expression, CNV and summed mutation burden. The pathway-based dataset Pancancer has two variants: PAN-2M uses mutation and CNV, whereas PAN-3M additionally includes gene expression. Additional datasets with full results are given in the supplement.

Models and baselines. We first compare bioMoR with three relevant configurations of transformers: *Vanilla* is the standard non-recursive Transformer. *Recursive* reuses one Transformer block for a fixed number of steps. *MoR* adds adaptive recursion using expert-choice or token-choice routing (Bae et al. 2025). **bioMoR** uses the same recursive backbone as MoR, but integrates biological knowledge at the embedding, attention, and router sites. We also compare with representative and state-of-the-art models, including Random Forest (Breiman 2001), Nearest-Centroid (Tibshirani et al. 2002), scBiGNN (Yang et al. 2023), CNN(ei) (Fu et al. 2020), MOGONET (Wang et al. 2021), Pathformer (Liu et al. 2024), Geneformer (Theodoris et al. 2023), scGPT (Cui et al. 2024), and scTransformer-style (Milia et al. 2026) baselines. Please see the supplementary material for details about these models.

Implementation and evaluation. Hyperparameters are fixed by modality, with no per-dataset tuning. The single-cell configuration follows the Genomap benchmark protocol (Islam and Xing 2023): AdamW, learning rate 10^{-3} , weight decay 10^{-5} , batch size 128, $d_{\text{model}}=96$, and $M=128$. The pathway-based multi-omics configuration follows PATH (Howlader, Islam, and Le 2026): AdamW, learning rate 3×10^{-4} , weight decay 10^{-2} , batch size 32, $d_{\text{model}}=128$, and $M=256$. Both use 10% linear warmup followed by cosine decay, gradient clipping at 1.0, dropout 0.1, at most 100 epochs, and patience 15. Results use paired 5-fold stratified cross-validation with seed 42; each fold holds out 20% for testing and 10% of training for validation. The default is $K=4$, with $K \in \{2, 3, 4\}$ tested in RQ4. Table 1 reports mean $\pm$ SD. See the supplement for the full hyperparameter configuration.

5 Results

We organize the results around four research questions. RQ1 evaluates predictive gains, RQ2 tests which biological-knowledge components matter, RQ3 measures whether the gains are computationally efficient, and RQ4 studies routing, depth, and model configurations.

RQ1: Does bioMoR improve predictive performance?

In RQ1, we compare bioMoR with a controlled architecture ladder across eight datasets to determine whether biology-guided routing improves predictive performance. Because these datasets are class-imbalanced, Table 1 reports macro-F1 and balanced accuracy. The architecture ladder consists of a vanilla Transformer, a fixed-depth recursive Transformer, and biology-agnostic MoR. To align effective depth with the base Transformer, denoted Vanilla, all recursive models in Table 1 use $K=4$. Both bioMoR routing variants outperform these baselines on average for both metrics.

Table 1: Macro-F1 and balanced accuracy (mean $\pm$ SD) from five-fold stratified cross-validation on eight benchmark datasets. All rows use the same folds and operating point; Recursive, MoR, and bioMoR use $K=4$. Avg. is the arithmetic mean across the eight displayed datasets. Underlining marks the top two results per dataset.

Metric	Model	Routing	Param	Single-cell					Pathway-based			Avg.
				Segerstolpe	Lung	T-cell	Muraro	Spleen	BLCA	PAN-2M	PAN-3M	
Macro-F1	Vanilla	–	300K	62.5 $\pm$ 11.2	72.5 $\pm$ 2.5	52.1 $\pm$ 3.5	80.2 $\pm$ 4.3	48.3 $\pm$ 3.7	40.6 $\pm$ 5.8	75.6 $\pm$ 3.5	95.2 $\pm$ 1.5	65.9
	Recursive	–	75K	61.4 $\pm$ 7.3	72.9 $\pm$ 2.9	50.5 $\pm$ 3.5	75.8 $\pm$ 4.1	49.4 $\pm$ 3.4	40.2 $\pm$ 1.1	72.3 $\pm$ 1.8	94.3 $\pm$ 1.1	64.6
	MoR	Expert	75K	64.3 $\pm$ 11.6	72.3 $\pm$ 2.7	50.9 $\pm$ 4.7	74.8 $\pm$ 4.7	53.2 $\pm$ 1.3	43.7 $\pm$ 2.8	75.5 $\pm$ 3.0	94.3 $\pm$ 1.3	66.1
		Token	75K	60.7 $\pm$ 5.8	73.3 $\pm$ 2.4	50.4 $\pm$ 2.2	81.6 $\pm$ 2.7	51.1 $\pm$ 3.1	40.6 $\pm$ 6.9	78.0 $\pm$ 4.6	95.3 $\pm$ 1.6	66.4
	bioMoR (ours)	Expert	75K	<u>72.2 $\pm$ 4.5</u>	<u>80.0 $\pm$ 1.2</u>	<u>70.0 $\pm$ 1.9</u>	<u>83.7 $\pm$ 4.9</u>	<u>59.7 $\pm$ 0.8</u>	<u>47.1 $\pm$ 7.1</u>	<u>85.8 $\pm$ 1.5</u>	<u>98.0 $\pm$ 0.4</u>	74.6
		Token	75K	<u>75.8 $\pm$ 2.8</u>	<u>79.7 $\pm$ 0.9</u>	<u>69.8 $\pm$ 1.2</u>	<u>81.9 $\pm$ 6.8</u>	<u>61.0 $\pm$ 0.6</u>	<u>39.5 $\pm$ 5.9</u>	<u>86.0 $\pm$ 1.9</u>	<u>96.4 $\pm$ 1.6</u>	73.8
Balanced accuracy	Vanilla	–	300K	71.7 $\pm$ 10.3	75.2 $\pm$ 1.5	55.2 $\pm$ 5.7	79.1 $\pm$ 3.3	55.3 $\pm$ 1.4	48.3 $\pm$ 2.8	74.1 $\pm$ 4.0	94.9 $\pm$ 1.9	69.2
	Recursive	–	75K	69.6 $\pm$ 9.1	75.3 $\pm$ 1.9	56.5 $\pm$ 4.6	77.8 $\pm$ 2.4	56.9 $\pm$ 1.5	48.5 $\pm$ 2.8	74.4 $\pm$ 1.6	93.6 $\pm$ 1.7	69.1
	MoR	Expert	75K	69.2 $\pm$ 6.1	76.5 $\pm$ 1.8	56.6 $\pm$ 6.1	75.5 $\pm$ 2.3	57.8 $\pm$ 2.5	<u>49.8 $\pm$ 5.0</u>	74.6 $\pm$ 3.1	93.7 $\pm$ 3.1	69.2
		Token	75K	66.4 $\pm$ 8.0	76.1 $\pm$ 2.2	56.6 $\pm$ 4.1	80.6 $\pm$ 8.5	56.5 $\pm$ 1.4	46.7 $\pm$ 7.8	73.5 $\pm$ 2.2	93.7 $\pm$ 4.6	68.8
	bioMoR (ours)	Expert	75K	<u>78.8 $\pm$ 4.9</u>	<u>80.4 $\pm$ 0.6</u>	<u>70.4 $\pm$ 1.6</u>	<u>88.9 $\pm$ 4.0</u>	<u>64.7 $\pm$ 1.6</u>	<u>48.7 $\pm$ 7.5</u>	<u>82.9 $\pm$ 3.3</u>	<u>95.4 $\pm$ 1.3</u>	76.3
		Token	75K	<u>73.3 $\pm$ 7.0</u>	<u>82.0 $\pm$ 1.5</u>	<u>71.3 $\pm$ 1.4</u>	<u>83.4 $\pm$ 3.7</u>	<u>64.1 $\pm$ 1.1</u>	48.3 $\pm$ 2.2	<u>83.4 $\pm$ 2.6</u>	<u>96.1 $\pm$ 1.9</u>	75.2

The best bioMoR configuration improves average macro-F1 by 8.2 percentage points and balanced accuracy by 7.1 percentage points over the strongest non-bioMoR baseline. Additional datasets and their performances are provided in the supplementary material.

Survival prediction. To test whether the architecture transfers beyond classification, we evaluate survival-risk prediction on four TCGA cohorts. Figure 2 compares bioMoR-Expert and bioMoR-Token with the Vanilla, Recursive, MoR-Expert, and MoR-Token baselines. The bioMoR variants are competitive across HNSC, UCEC, COAD, and KIRP, showing that biology-guided adaptive recursion also transfers to a time-to-event endpoint.

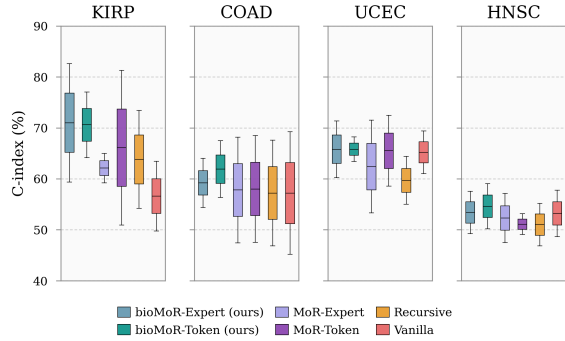


Figure 2: **Survival prediction across TCGA cohorts.** Fold-level C-index distributions for HNSC, UCEC, COAD, and KIRP. Higher values indicate better concordance between predicted risk and observed survival outcomes.

Beyond predictive performance, we next examine whether these gains are reflected in the learned representations. Figure 3 visualizes the frozen embeddings on Segerstolpe using UMAP (McInnes et al. 2018) and evaluates their linear separability with a linear probe (Alain and Bengio 2016). As shown in Figure 3, bioMoR forms cleaner cell-type groups in the frozen embedding space, and the linear probe performs best on these embeddings. This suggests that bioMoR im-

proves the learned representation rather than only the final prediction layer. Additional visualizations are provided in the supplement.

Finally, to assess whether the gains extend beyond our internal architecture ladder, Figure 4 compares bioMoR with dataset-appropriate external state-of-the-art baselines on PAN-2M and Lung as representative datasets. The full set of baselines is described under *Models and baselines*. bioMoR obtains the highest mean macro-F1 in both panels: 0.86 ± 0.01 on PAN-2M, compared with 0.84 ± 0.01 for scBiGNN, and 0.80 ± 0.01 on Lung, compared with 0.78 ± 0.03 for TabNet. Additional results for external baselines are provided in the supplement.

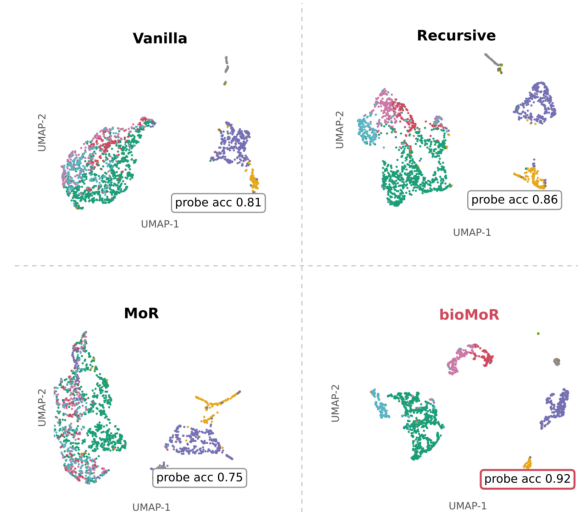


Figure 3: **Frozen-embedding linear probe on Segerstolpe dataset.** UMAP visualization of penultimate per-cell embeddings for Vanilla, Recursive, MoR, and bioMoR, colored by cell type.

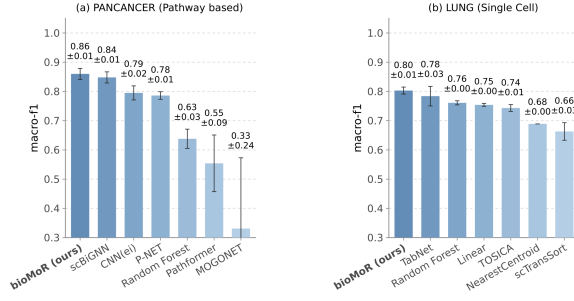


Figure 4: **Comparison with external baselines.** Mean macro-F1 $\pm$ SD for bioMoR and dataset-appropriate baselines on (a) pathway-based PAN-2M and (b) single-cell Lung under the matched five-fold evaluation setting.

Ablation Results

RQ2: In which component(s), the biological knowledge is most useful? To evaluate the contribution of each biological injection site, we conduct an ablation study on the token embedding, attention bias, and router. We ran all the 8 datasets, and due to the space, we showed results for four representative datasets in Figure 5. Our results indicate that integrating biological knowledge at all three sites achieves the highest macro-F1. Among the single-site variants, embedding injection performs best, followed by router-only and attention-bias-only injection. Overall, these results show that the three injection sites provide complementary information, with the strongest performance obtained when all three are used together.

Efficiency

RQ3: What are the training and FLOP gains? RQ3 asks whether bioMoR improves training behavior while reducing computational cost. We evaluate computational efficiency across all datasets and provide the complete results in the supplementary material. Due to the main-paper space limit, we present T-cell as a representative dataset from Table 1. With expert-choice routing at $K=4$, we compare learning curves and checkpoint-selected test performance across the internal architecture ladder (Figure 6a–c). In panel (a), bioMoR reduces the training loss faster and reaches the lowest final value. In panel (b), it achieves nearly 70% validation macro-F1, whereas the baselines remain near 40–50%. Panel (c) confirms this advantage under both checkpoint-selection rules, with bioMoR achieving the highest test macro-F1 of 69.9%.

Panel (d) compares predictive performance and computational cost. Each circle represents one model at a specific recursion depth. Models closer to the upper-left are preferable because they achieve higher macro-F1 with fewer FLOPs. bioMoR consistently maintains approximately 69–70% macro-F1 across all recursion depths while requiring fewer FLOPs than Vanilla. At $K=2$, bioMoR uses 58% fewer FLOPs than Vanilla.

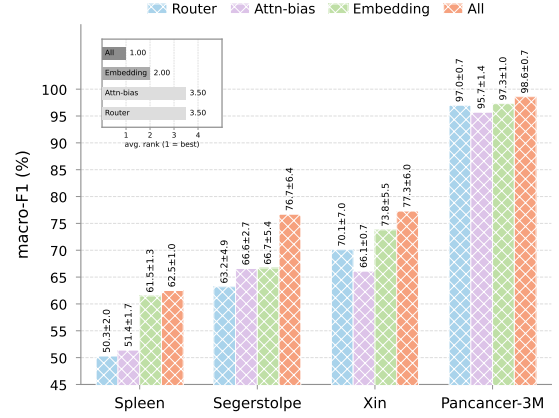


Figure 5: **Injection-site ablation.** Macro-F1 across Spleen, Segerstolpe, Xin, and Pancancer-3M with biological-knowledge injection at the embedding, attention-bias, router, or all three sites. All experiments use expert-choice routing with $K=4$. The inset shows the average rank across datasets (1 is best).

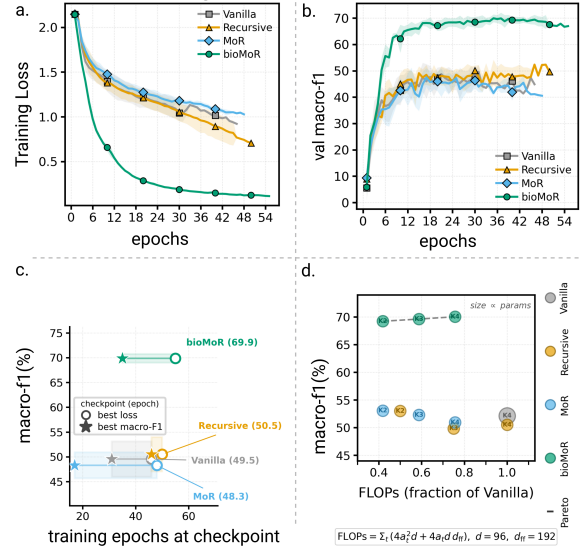


Figure 6: **Training and computational efficiency on T-cell.** (a) Training loss and (b) validation macro-F1 over epochs. (c) Test macro-F1 and training epoch at checkpoints selected by lowest validation loss (○) or highest validation macro-F1 (★). (d) Test macro-F1 versus FLOPs for $K=2, 3, 4$; bubble size represents parameter count.

Configuration Studies

RQ4: Which configuration choices matter? RQ4 examines whether bioMoR depends on a specific configuration. We evaluate all datasets and report the complete results in the supplement; due to space limits, Table 2 shows T-cell as a representative case. bioMoR remains strong under both routing policies and across $K=2, 3, 4$, while Vanilla, Recursive, and MoR perform substantially worse. Thus, biological guidance drives the main gain, while routing policy and depth adjust the accuracy–efficiency trade-off.

Table 2: **RQ4 T-cell configuration slice.** T-cell macro-F1 from Supplementary Table 1. Each cell reports normalized FLOPs / macro-F1 (mean $\pm$ SD).

Model	Routing	Normalized FLOPs / macro-F1		
		$K=2$	$K=3$	$K=4$
Vanilla	–	–	–	1.00/52.1 $\pm$ 3.5
Recursive	–	0.50/53.0 $\pm$ 2.0	0.75/49.8 $\pm$ 4.4	1.00/50.5 $\pm$ 3.5
MoR	Expert	0.42/50.6 $\pm$ 4.9	0.59/52.3 $\pm$ 2.4	0.76/50.9 $\pm$ 4.7
MoR	Token	0.42/50.7 $\pm$ 2.2	0.52/55.8 $\pm$ 3.0	0.56/50.4 $\pm$ 2.2
bioMoR	Expert	0.42/69.2 $\pm$ 0.7	0.59/69.6 $\pm$ 0.6	0.76/70.0 $\pm$ 1.9
bioMoR	Token	0.42/68.6 $\pm$ 1.7	0.52/68.6 $\pm$ 0.9	0.56/69.8 $\pm$ 1.2

Tables 3 and 4 examine the marker budget M and model width d , respectively. Mean macro-F1 remains within 71.8–73.5 as M increases from 128 to 2048: T-cell and Spleen improve, Lung remains stable, and Muraro and Segerstolpe show no consistent trend. As d increases from 96 to 352, mean macro-F1 rises from 70.5 to 72.7, although the change is not monotonic; Lung, T-cell, and Spleen remain stable, Muraro generally improves, and Segerstolpe, BLCA, and PAN-2M vary across settings. Overall, bioMoR’s improvement does not depend on a specific configuration; the performance gain is primarily driven by biological knowledge.

Table 3: **RQ4 marker-budget headroom.** bioMoR macro-F1 (mean $\pm$ SD) across the eight datasets shown in Table 1.

Dataset	$M=128$	$M=256$	$M=512$	$M=1024$	$M=2048$
Lung	79.1 $\pm$ 1.2	80.2 $\pm$ 0.8	79.8 $\pm$ 1.9	81.3 $\pm$ 2.0	81.1 $\pm$ 1.9
Muraro	75.3 $\pm$ 6.4	72.4 $\pm$ 2.7	68.3 $\pm$ 3.3	75.7 $\pm$ 4.5	73.8 $\pm$ 6.0
Segerstolpe	71.4 $\pm$ 5.3	70.3 $\pm$ 4.1	73.8 $\pm$ 5.4	69.4 $\pm$ 14.7	68.3 $\pm$ 3.3
Spleen	58.2 $\pm$ 0.9	58.6 $\pm$ 1.6	59.9 $\pm$ 0.8	61.1 $\pm$ 2.3	61.6 $\pm$ 4.0
T-cell	68.9 $\pm$ 1.0	69.9 $\pm$ 0.3	71.5 $\pm$ 0.4	71.8 $\pm$ 2.3	73.0 $\pm$ 0.3
BLCA	44.9 $\pm$ 7.0	41.5 $\pm$ 6.7	42.6 $\pm$ 10.0	45.6 $\pm$ 4.9	44.6 $\pm$ 6.4
PAN-2M	85.5 $\pm$ 1.9	85.0 $\pm$ 0.9	85.7 $\pm$ 2.1	86.3 $\pm$ 1.3	84.4 $\pm$ 0.9
PAN-3M	96.7 $\pm$ 1.4	96.1 $\pm$ 0.7	96.3 $\pm$ 0.5	96.4 $\pm$ 1.3	95.5 $\pm$ 0.8
Mean	72.5	71.8	72.2	73.5	72.8

Table 4: **RQ4 model-width sweep (5-fold CV).** bioMoR macro-F1 (mean $\pm$ SD over the unified five folds) across $d_{\text{model}} \in \{96, 136, 192, 272, 352\}$ at fixed marker budget $M=256$, on five single-cell and three multi-omics datasets.

Dataset	$d=96$	$d=136$	$d=192$	$d=272$	$d=352$
Lung	80.2 $\pm$ 1.4	79.6 $\pm$ 0.7	80.6 $\pm$ 1.8	81.6 $\pm$ 0.9	80.4 $\pm$ 1.0
Muraro	79.7 $\pm$ 2.3	85.4 $\pm$ 5.9	82.6 $\pm$ 6.6	87.4 $\pm$ 5.6	88.6 $\pm$ 5.6
Segerstolpe	76.0 $\pm$ 3.4	75.2 $\pm$ 6.7	79.9 $\pm$ 4.2	74.8 $\pm$ 6.6	80.5 $\pm$ 6.9
Spleen	60.6 $\pm$ 0.6	60.3 $\pm$ 1.6	61.7 $\pm$ 1.3	60.3 $\pm$ 0.4	60.9 $\pm$ 0.9
T-cell	69.5 $\pm$ 1.0	69.2 $\pm$ 2.0	70.1 $\pm$ 1.3	69.7 $\pm$ 0.9	69.2 $\pm$ 1.9
BLCA	42.2 $\pm$ 2.1	43.1 $\pm$ 8.4	42.8 $\pm$ 9.3	44.6 $\pm$ 5.3	42.3 $\pm$ 3.6
PAN-2M	85.7 $\pm$ 0.5	86.4 $\pm$ 1.9	86.9 $\pm$ 1.6	84.6 $\pm$ 2.5	87.1 $\pm$ 1.6
PAN-3M	96.7 $\pm$ 1.4	96.1 $\pm$ 0.7	96.3 $\pm$ 0.5	96.4 $\pm$ 1.3	95.5 $\pm$ 0.8
Mean	73.8	74.4	75.1	74.9	75.6

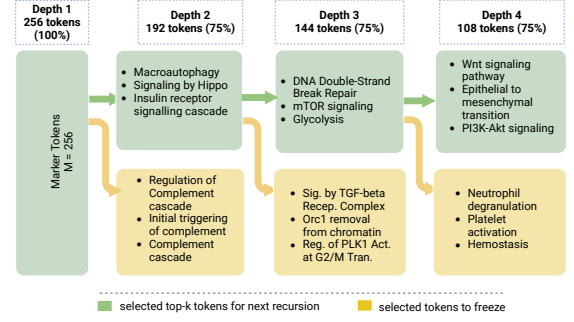


Figure 7: **Pathway-prioritization case study.** Green boxes give examples of pathway tokens assigned to receive deeper computation; yellow boxes give examples that stop at the corresponding depth and retain their current representations.

Biological Case Study: Pathways Prioritized for Deeper Computation

We use PAN-2M to examine which pathway tokens bioMoR processes more deeply for localized-versus-metastatic cancer classification. Because each token represents a named biological pathway, the routing decisions can be compared with existing cancer studies.

At each recursion, the router sends the top 75% of active pathways to the next depth, while the remaining 25% stop. Thus, pathways reaching deeper levels receive more computation. In Figure 7, green boxes indicate continuing pathways and yellow boxes indicate stopped pathways.

Pathways receiving the deepest computation include Wnt signaling, epithelial-to-mesenchymal transition (EMT), and PI3K–Akt signaling, all of which are linked to cancer invasion or metastasis (Nguyen et al. 2009; Grasset et al. 2022; Thibault et al. 2021). This agreement with prior studies indicates that bioMoR prioritizes pathways known to be relevant to metastatic cancer.

6 Conclusion

We developed bioMoR, a biological-knowledge-guided Mixture-of-Recursions framework and, to the best of our knowledge, the first application of MoR to genomic learning. We integrate biological knowledge at three sites: to refine token embeddings, bias self-attention, and route each token to an appropriate recursion depth. This design directly addresses a central computational mismatch in omics modeling: most features should not require the same amount of deep computation, and the decision about where to spend computation should be informed by molecular structure rather than by token embeddings alone. Our evaluation demonstrates that bioMoR significantly improves predictive performance while being more computationally efficient, as shown by fewer model parameters and reduced FLOPs. These improvements are consistent across diverse genomic datasets and the extensive configurations we evaluated. Overall, our results suggest that biological structure is useful not only as knowledge for representation learning, but also as a control signal for adaptive computation in genomic models.

Acknowledgments

This work is partially supported by the National Science Foundation under Grant No. 2152117. The research reported in this paper is partially supported by the HPC@ISU equipment at Iowa State University, some of which has been purchased through funding provided by NSF under MRI grants number 1726447 and MRI2018594.

References

- Alain, G., and Bengio, Y. 2016. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*.
- Bae, S.; Kim, Y.; Bayat, R.; Kim, S.; Ha, J.; et al. 2025. Mixture-of-recursions: Learning dynamic recursive depths for adaptive token-level computation. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Breiman, L. 2001. Random forests. *Machine Learning* 45:5–32.
- Choromanski, K.; Likhoshesterov, V.; Dohan, D.; Song, X.; Gane, A.; Sarlos, T.; et al. 2021. Rethinking attention with performers. In *International Conference on Learning Representations (ICLR)*.
- Cui, H.; Wang, C.; Maan, H.; Pang, K.; Luo, F.; Duan, N.; and Wang, B. 2024. scgpt: Toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* 21:1470–1480.
- Dehghani, M.; Gouws, S.; Vinyals, O.; Uszkoreit, J.; and Kaiser, L. 2019. Universal transformers. In *International Conference on Learning Representations (ICLR)*.
- Elmarakeby, H. A.; Hwang, J.; Arafeh, R.; et al. 2021. Biologically informed deep neural network for prostate cancer discovery. *Nature* 598:348–352.
- Feuk, L.; Carson, A. R.; and Scherer, S. W. 2006. Structural variation in the human genome. *Nature Reviews Genetics* 7(2):85–97.
- Franzén, O.; Gan, L.-M.; and Björkegren, J. L. 2019. Panglaodb: A web server for exploration of mouse and human single-cell RNA sequencing data. *Database* 2019:baz046.
- Fu, Y.; Xu, J.; Tang, Z.; et al. 2020. A gene prioritization method based on a swine multi-omics knowledgebase and a deep learning model. *Communications Biology* 3:502.
- Gillespie, M.; Jassal, B.; Stephan, R.; Milacic, M.; Rothfels, K.; Senff-Ribeiro, A.; Griss, J.; et al. 2022. The reactome pathway knowledgebase 2022. *Nucleic Acids Research* 50(D1):D687–D692.
- Goldman, M. J.; Craft, B.; Hastie, M.; Repečka, K.; McDade, F.; Kamath, A.; Banerjee, A.; Luo, Y.; Rogers, D.; Brooks, A. N.; Zhu, J.; and Haussler, D. 2020. Visualizing and interpreting cancer genomics data via the Xena platform. *Nature Biotechnology* 38(6):675–678.
- Grasset, E. M.; Dunworth, M.; Sharma, G.; Loth, M.; Tandurella, J.; Cimino-Mathews, A.; Gentz, M.; Bracht, S.; Haynes, M.; Fertig, E. J.; and Ewald, A. J. 2022. Triple-negative breast cancer metastasis involves complex epithelial-mesenchymal transition dynamics and requires vimentin. *Science Translational Medicine* 14(656):eabn7571.
- Graves, A. 2016. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.
- Hao, M.; Gong, J.; Zeng, X.; Liu, C.; Guo, Y.; Cheng, X.; Wang, T.; Ma, J.; Zhang, X.; and Song, L. 2024. Large-scale foundation model on single-cell transcriptomics. *Nature Methods* 21:1481–1491.
- Harrell, F. E. J.; Lee, K. L.; Califf, R. M.; Pryor, D. B.; and Rosati, R. A. 1984. Regression modelling strategies for improved prognostic prediction. *Statistics in Medicine* 3(2):143–152.
- Howlader, K.; Islam, M. T.; and Le, W. 2026. Graph transformer-based pathway embedding for cancer prognosis. *arXiv preprint arXiv:2604.16685*.
- Hu, C.; Li, T.; Xu, Y.; Zhang, X.; Li, F.; Bai, J.; Chen, J.; Jiang, W.; Yang, K.; Ou, Q.; Li, X.; Wang, P.; and Zhang, Y. 2023. Cellmarker 2.0: An updated database of manually curated cell markers in human/mouse and web tools based on scRNA-seq data. *Nucleic Acids Research* 51(D1):D870–D876.
- Islam, M. T., and Xing, L. 2023. Cartography of genomic interactions enables deep analysis of single-cell expression data. *Nature Communications* 14:679.
- Lan, Z.; Chen, M.; Goodman, S.; Gimpel, K.; Sharma, P.; and Soricut, R. 2020. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR)*.
- Liu, X.; Tao, Y.; Cai, Z.; Bao, P.; Ma, H.; Li, K.; Li, M.; Zhu, Y.; and Lu, Z. J. 2024. Pathformer: A biological pathway informed transformer for disease diagnosis and prognosis using multi-omics data. *Bioinformatics* 40(5):btac316.
- McInnes, L.; Healy, J.; Saul, N.; and Großberger, L. 2018. UMAP: Uniform manifold approximation and projection. *Journal of Open Source Software* 3(29):861.
- Milia, M.; Tshimanga, L. F.; Mueller, H.; Atzori, M.; and Di Camillo, B. 2026. Integrating gene regulatory priors into transformer attention with scTransformer for interpretable scRNA-seq analysis. *arXiv preprint arXiv:2606.09558*.
- Nguyen, D. X.; Chiang, A. C.; Zhang, X. H.-F.; Kim, J. Y.; Kris, M. G.; Ladanyi, M.; Gerald, W. L.; and Massagué, J. 2009. WNT/TCF signaling through LEF1 and HOXB9 mediates lung adenocarcinoma metastasis. *Cell* 138(1):51–62.
- Raposo, D.; Ritter, S.; Richards, B.; Lillicrap, T.; Humphreys, P. C.; and Santoro, A. 2024. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*.
- Schwartz, R.; Dodge, J.; Smith, N. A.; and Etzioni, O. 2020. Green AI. *Communications of the ACM* 63(12):54–63.
- Shazeer, N.; Mirhoseini, A.; Maziarz, K.; Davis, A.; Le, Q.; Hinton, G.; and Dean, J. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.
- The Cancer Genome Atlas Research Network; Weinstein, J. N.; Collisson, E. A.; Mills, G. B.; Shaw, K. R. M.; Ozenberger, B. A.; Ellrott, K.; Shmulevich, I.; Sander, C.; and Stuart, J. M. 2013. The cancer genome atlas pan-cancer analysis project. *Nature Genetics* 45:1113–1120.
- Theodoris, C. V.; Xiao, L.; Chopra, A.; Chaffin, M. D.; Al Sayed, Z. R.; Hill, M. C.; Mantineo, H.; Brydon, E. M.; Zeng, Z.; Liu, X. S.; and Ellinor, P. T. 2023. Transfer learning enables predictions in network biology. *Nature* 618:616–624.
- Thibault, B.; Ramos-Delgado, F.; Pons-Tostivint, E.; Theriville, N.; Cintas, C.; Arcucci, S.; Cassant-Sourdy, S.; Reyes-Castellanos, G.; Tosolini, M.; Villard, A. V.; et al. 2021. Pancreatic cancer intrinsic PI3K α activity accelerates metastasis and rewires macrophage component. *EMBO Molecular Medicine* 13(7):e13502.
- Tibshirani, R.; Hastie, T.; Narasimhan, B.; and Chu, G. 2002. Diagnosis of multiple cancer types by shrunken centroids of gene expression. *Proceedings of the National Academy of Sciences* 99(10):6567–6572.
- Vogelstein, B.; Papadopoulos, N.; Velculescu, V. E.; Zhou, S.; Diaz, L. A. J.; and Kinzler, K. W. 2013. Cancer genome landscapes. *Science* 339(6127):1546–1558.

- Wang, S.; Li, B. Z.; Khabsa, M.; Fang, H.; and Ma, H. 2020. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*.
- Wang, T.; Shao, W.; Huang, Z.; Tang, H.; Zhang, J.; Ding, Z.; and Huang, K. 2021. MOGONET integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification. *Nature Communications* 12:3445.
- Wang, Z.; Gerstein, M.; and Snyder, M. 2009. RNA-Seq: A revolutionary tool for transcriptomics. *Nature Reviews Genetics* 10(1):57–63.
- Xiong, Y.; Zeng, Z.; Chakraborty, R.; Tan, M.; Fung, G.; Li, Y.; and Singh, V. 2021. Nyströmformer: A nyström-based algorithm for approximating self-attention. In *AAAI Conference on Artificial Intelligence*.
- Yang, F.; Wang, W.; Wang, F.; Fang, Y.; Tang, D.; Huang, J.; Lu, H.; and Yao, J. 2022. scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data. *Nature Machine Intelligence* 4(10):852–866.
- Yang, R.; Dai, W.; Li, C.; Zou, J.; Wu, D.; and Xiong, H. 2023. scBiGNN: Bilevel graph representation learning for cell type classification from single-cell RNA sequencing data. *arXiv preprint arXiv:2312.10310*.
- Zhou, Y.; Lei, T.; Liu, H.; Du, N.; Huang, Y.; Zhao, V.; Dai, A. M.; Chen, Z.; Le, Q. V.; and Laudon, J. 2022. Mixture-of-experts with expert choice routing. In *Advances in Neural Information Processing Systems*, volume 35.

Supplementary Material

bioMoR: Biology-Guided Mixture-of-Recursions for Effective Genomic Learning

Koushik Howlader¹, Tirtho Roy¹, Md Tauhidul Islam², Wei Le¹

¹Iowa State University, Iowa, USA

²Stanford University, CA, USA

Appendix A: Extended Related Work

Genomic representation learning. Genomic samples contain thousands of correlated and often sparse molecular measurements. Single-cell Transformers such as scBERT (Yang et al. 2022), Geneformer (Theodoris et al. 2023), and scGPT (Cui et al. 2024) show that large-scale pre-training can support cell annotation, perturbation analysis, and transfer to network-biology tasks; scFoundation further studies scaling to larger cell collections (Hao et al. 2024). These models establish the value of learned gene representations, but their standard backbones normally apply the same sequence of layers to every retained gene. Related feature-selection methods focus computation on highly variable or marker genes (Luecken and Theis 2019; Ianevski, Giri, and Aittokallio 2022), while Genomap exploits structured gene relationships to construct compact single-cell representations (Islam and Xing 2023). Selection, however, does not determine how deeply each retained feature should be processed. bioMoR separates these decisions: it first forms interpretable marker or pathway tokens and then assigns token-specific recursive depth.

Biological priors in omics models. Biological structure enters omics models through curated hierarchies, pathways, and interaction graphs. P-NET constrains a cancer model by gene/pathway organization (Elmarakeby et al. 2021); TOSICA uses pathway-informed attention for interpretable cell annotation (Chen et al. 2023); Pathformer models pathway interactions in multi-omics data (Liu et al. 2024); and MOGONET and scBiGNN use graph structure for multi-omics and single-cell prediction (Wang et al. 2021; Yang et al. 2023). These studies show that molecular relations improve representation quality and interpretability. Most use biology to define connectivity, aggregate features, or bias attention, rather than to control whether a token is processed again. bioMoR uses one knowledge source at three sites: graph smoothing refines embeddings, structural bias guides attention, and a neighborhood-aware router determines recursion depth.

Parameter-efficient depth and adaptive computation.

Universal Transformers repeatedly apply one transition block (Dehghani et al. 2019), while ALBERT shows that cross-layer sharing can reduce parameters (Lan et al. 2020); both nevertheless use fixed computation. Adaptive Computation Time learns when a recurrent state should halt (Graves 2016), sparsely gated mixture-of-experts routes tokens among experts (Shazeer et al. 2017), and Mixture-of-Depths allocates different amounts of layer computation under a budget (Raposo et al. 2024). Mixture-of-Recursions combines adaptive routing with a shared recursive block, decoupling maximum depth from parameter count (Bae et al. 2025). These routers are general-purpose and must infer biological relationships only from task data.

Positioning of bioMoR. bioMoR connects biology-informed genomic learning with adaptive recursion. It is not a replacement for foundation-model pretraining or a new pathway database; it adds a biology-guided computation policy to a compact Transformer. Compared with fixed biological architectures, it allocates unequal depth across tokens. Compared with biology-agnostic MoR, it uses gene co-expression or Reactome relations to inform representations, attention, and routing while preserving a no-op path for the biological correction. Thus, bioMoR addresses the main paper’s central mismatch: genomic tokens differ in both predictive relevance and the relational computation needed to resolve their contribution.

Appendix B: Dataset and Baseline Details

Dataset Selection

The main paper reports a representative eight-dataset classification subset. The complete classification suite contains 14 datasets: 8 Genomap single-cell datasets (Islam and Xing 2023), 3 Reactome multi-omics cohorts (Gillespie et al. 2022), and 3 pan-cancer TCGA settings (The Cancer Genome Atlas Research Network et al. 2013). Table 1 summarises the dataset name, sample count, feature type, and classification type.

Dataset	Samples	Feature type	Classification type
<i>Single-cell genomap datasets</i>			
Baron	8,569	scRNA-seq Expression	Multi-class (13 classes)
Lung	57,020	scRNA-seq Expression	Multi-class (25 classes)
Muraro	2,122	scRNA-seq Expression	Multi-class (9 classes)
Oesophagus	87,947	scRNA-seq Expression	Multi-class (18 classes)
Segerstolpe	2,127	scRNA-seq Expression	Multi-class (11 classes)
Spleen	94,129	scRNA-seq Expression	Multi-class (28 classes)
T-cell	24,007	scRNA-seq Expression	Multi-class (7 classes)
Xin	1,492	scRNA-seq Expression	Multi-class (4 classes)
<i>Cancer pathway-token datasets</i>			
Prostate	1,011	Mutation+CNV+Expression	Binary classification
BLCA (bladder)	404	Mutation+CNV+Expression	Binary classification
STAD (stomach)	414	Mutation+CNV+Expression	Binary classification
Pan-cancer mutation+CNV	8,893	CNV+Mutation	Binary classification
Pan-cancer expression	8,586	Expression	Binary classification
Pan-cancer tri-modal	8,586	CNV+Mutation+Expression	Binary classification

Table 1: Dataset summary. Cancer datasets use Reactome pathway-token inputs; the pan-cancer cohorts evaluate three feature views: CNV+mutation, expression only, and CNV+mutation+expression.

Baseline Selection

Model or baseline	Role in comparison	Reference
Vanilla Transformer	Non-recursive internal baseline	(Vaswani et al. 2017)
Recursive Transformer	Weight-shared fixed-depth baseline	(Dehghani et al. 2019; Lan et al. 2020)
MoR	Biology-agnostic adaptive-recursion baseline	(Bae et al. 2025)
Random Forest	Classical tree-based classifier	(Breiman 2001)
Nearest-Centroid	Classical centroid-based classifier	(Tibshirani et al. 2002)
TabNet	Attentive tabular-learning baseline	(Arik and Pfister 2021)
TOSICA	Pathway-aware single-cell Transformer	(Chen et al. 2023)
scBiGNN	Single-cell graph neural baseline	(Yang et al. 2023)
MOGONET	Multi-omics graph neural baseline	(Wang et al. 2021)
Pathformer	Pathway-informed transformer baseline	(Liu et al. 2024)
Geneformer	Gene-vocabulary foundation model baseline	(Theodoris et al. 2023)
scGPT	Single-cell foundation model baseline	(Cui et al. 2024)
scTransformer-style	Transformer with gene-regulatory priors	(Milia et al. 2026)
CNN(ei)	Expression-image convolutional baseline	(Fu et al. 2020)

Table 2: Model and baseline references used in the experimental comparisons. Internal baselines test architectural changes under the same bioMoR input pipeline; external baselines are included when their input assumptions match the corresponding dataset type.

Appendix C: Experimental Setup

Hyperparameter configuration. Fixed training configuration (no per-dataset tuning); each modality uses one configuration held fixed across all its datasets and variants. *LR schedule*: linear warmup over the first 10% of epochs (1% $\rightarrow$ 100% of the base rate), then cosine annealing. *Early stopping* on validation macro-F1. *Protocol*: 5-fold stratified cross-validation (seed 42; 20% test, 10%-of-train validation), per-gene z -scoring fit on the train fold only; class-weighted cross-entropy loss. M counts learnable marker tokens (single-cell) or fixed Reactome pathway tokens (multi-omics).

Setting	Single-cell	Multi-omics
Optimiser	AdamW	AdamW
Base learning rate	10^{-3}	3×10^{-4}
Weight decay	10^{-5}	10^{-2}
LR schedule	warmup $\rightarrow$ cosine	warmup $\rightarrow$ cosine
Gradient clip (ℓ_2)	1.0	1.0
Dropout	0.1	0.1
Batch size	128	32
Max epochs	100	100
Early-stop patience	15	15
Width d_{model}	96	128
FFN width d_{ff}	192	256
Token budget M	128	256
Recursion depth K	4	4
CV folds / seed	5 / 42	5 / 42

Appendix D: Extended Method

bioMoR

Overview and notation. For one sample, x denotes the measured genomic input: a gene-expression vector for single-cell data or aligned mutation, copy-number, and expression channels for multi-omics data. Tokenization converts x into $T = [t_1, \dots, t_M]^\top \in \mathbb{R}^{M \times d}$, where M is the number of marker-gene or pathway tokens, d is the hidden width, and t_m is the initial representation of token m . The symbol K denotes the largest permitted recursion depth. At recursion step t , $H^{(t)} \in \mathbb{R}^{M \times d}$ stores the current token states, while $\mathcal{A}_t \subseteq \{1, \dots, M\}$ contains only the tokens that remain active. A token removed from $\mathcal{A}_t$ keeps its latest state, so routing changes computation without deleting its information from the final pooled representation.

Algorithm 1: bioMoR forward pass.

Input: x , biological knowledge B
token budget M , max depth K
 $T \leftarrow$ marker-gene or pathway tokens from x
 $A \leftarrow$ row-normalized fixed knowledge matrix from B
 $H \leftarrow (1 - \lambda)T + \lambda AT$; $\mathcal{A} \leftarrow \{1, \dots, M\}$
for $t = 1, \dots, K$ **while** $\mathcal{A} \neq \emptyset$ **do**
 $H_{\mathcal{A}} \leftarrow f_\theta(H_{\mathcal{A}}; A)$ *reuse the same block*
 $r_{\mathcal{A}} \leftarrow$ biology-aware router scores
 $\mathcal{A}_{\text{next}} \leftarrow$ tokens routed to the next step
tokens in $\mathcal{A} \setminus \mathcal{A}_{\text{next}}$ stop and keep current states
 $\mathcal{A} \leftarrow \mathcal{A}_{\text{next}}$ *return kept tokens to recursion*
end for
 $\hat{y} \leftarrow$ classifier(pool(H))
Train: minimize class-weighted CE plus auxiliary losses
Output: $\hat{y}$, markers/pathways, recursion depths d_m

Reading Algorithm 1. B is fixed biological knowledge and A is its normalized form. The shared Transformer f_θ has trainable parameters θ and is reused at every step, so increasing K increases possible computation but not the number of block parameters. The score vector $r_{\mathcal{A}}$ ranks or gates the active tokens; $\mathcal{A}_{\text{next}}$ is the subset sent through another application of f_θ . After routing stops, pool(H) combines all final token states, the classifier produces the predicted label distribution $\hat{y}$, and $d_m \in \{1, \dots, K\}$ records how many recursive updates token m received. Consequently, d_m is both a compute-allocation variable and an interpretable summary of which marker or pathway required more refinement.

Biological-knowledge construction. Following Genomap (Islam and Xing 2023), we construct a gene *co-expression* matrix that summarizes pairwise gene relationships. Its entries are partial correlations computed from the inverse covariance of the gene-expression matrix and restricted to the selected markers; this matrix forms the single-cell biological knowledge B . For multi-omics data, we use curated Reactome pathway relationships and treat them as undirected: $B_{ij} = B_{ji} = 1$ when pathways i and j are related, and both entries are zero otherwise (Gillespie et al. 2022).

Let D be the diagonal matrix of row sums of B . We use the fixed, row-normalized matrix

$$A = D^{-1}B. \quad (1)$$

B_{ij} measures or indicates the biological relation between tokens i and j ; $D_{ii} = \sum_j B_{ij}$ is token i 's total connection weight; and $A_{ij} = B_{ij}/D_{ii}$ is the fraction of its neighborhood assigned to token j . Thus, row i of AH is a weighted average of the current states of token i 's biological neighbors. Row normalization prevents highly connected genes or pathways from producing a larger message merely because they have more neighbors. Isolated tokens, if present, receive no neighbor message and therefore rely on their own representation through the residual mixture in Site 1.

The same biological knowledge is used for embedding smoothing (Site 1), attention bias (Site 2), and depth routing (Site 3).

Site 1: biology-guided embedding smoothing. Before recursion, we smooth the tokens using the biological-knowledge graph A , so that information of biologically related tokens is also integrated into the embedding of the token:

$$\tilde{T} = (1 - \lambda)T + \lambda AT, \quad (2)$$

where λ is learned. This denoises each marker/pathway token toward its biological neighbours before the repeated recursive updates.

Here, $\tilde{T}$ is the representation passed to the first recursive step, AT is the neighborhood-averaged representation, and λ controls the mixture. When $\lambda = 0$, Site 1 reduces to the original token representation; larger positive values give biological neighbors more influence. This site can improve robustness when related genes or pathways carry complementary signal, but an excessively strong graph contribution can blur genuinely distinct tokens. Learning λ lets validation loss determine the useful balance rather than fixing it manually.

Site 2: attention bias. Inside the recursive block, the *same* biological-knowledge graph A biases self-attention:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + \lambda_{\text{attn}}\mathbf{1}[A > 0]\right)V. \quad (3)$$

This encourages biologically related tokens to exchange information during refinement.

Q , K , and V are the query, key, and value matrices computed from the current token states; the K in this equation means the *key matrix*, not maximum recursion depth. The factor $\sqrt{d}$ keeps dot-product magnitudes stable as the hidden width changes. The binary mask $\mathbf{1}[A > 0]$ equals one for a biological edge and zero otherwise, while λ_{attn} is the learned strength of the added logit bias. Therefore, the graph does not force attention: it raises the prior logit of related pairs, and the data-dependent term $QK^\top$ can still override it. This can concentrate information exchange on plausible relationships without preventing the model from learning task-specific nonlocal interactions.

Site 3: biology-aware routing. The original MoR router uses the current token representation $h_m^{(t)}$ to decide whether token m should continue to the next recursion step. bioMoR

keeps this original score and adds a learned correction based on the token’s biological neighborhood:

$$\hat{r}_m^{(t)} = \frac{w_r^\top h_m^{(t)}}{\tau_r} + \sigma(\gamma_t) \phi_\psi \left([AH^{(t)}, H^{(t)} - AH^{(t)}]_m \right). \quad (4)$$

The first term is the original MoR routing score. In the second term, $AH^{(t)}$ summarizes information from biologically related tokens, while $H^{(t)} - AH^{(t)}$ shows how the token differs from its biological neighborhood. The small network ϕ_ψ uses these two signals to adjust the routing score, and $\sigma(\gamma_t)$ controls how much correction is added at recursion step t . Because ϕ_ψ is initialized to output zero, the correction is zero at the start of training and bioMoR initially behaves like biology-agnostic MoR. During training, the correction becomes nonzero when biological information helps reduce the task loss; otherwise, it can remain close to zero. This design prevents noisy biological knowledge from affecting routing at the beginning while allowing the model to use it when helpful.

More explicitly, m indexes a token and t indexes a recursion step; $h_m^{(t)} \in \mathbb{R}^d$ is row m of $H^{(t)}$; $w_r \in \mathbb{R}^d$ is the base-router weight vector; and $\tau_r > 0$ is a temperature controlling how sharply router scores separate. The vector $[\cdot, \cdot]_m$ concatenates token m ’s neighborhood mean, $[AH^{(t)}]_m$, with its neighborhood residual, $[H^{(t)} - AH^{(t)}]_m$. The small network ϕ_ψ , with parameters ψ , maps this evidence to one scalar correction. The sigmoid $\sigma(\gamma_t) \in (0, 1)$ is a learned step-specific gate. A higher corrected score makes another recursive update more likely under the chosen routing policy. Performance can improve when the correction spends depth on tokens whose local biological context is informative; if the graph is unhelpful, zero initialization and the learned gate provide a route back toward ordinary MoR behavior.

Training and efficiency. The main task loss is class-weighted cross-entropy. During training we minimize

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{suf}} + \beta \mathcal{L}_{\text{div}} + \gamma \mathcal{L}_{\text{cmp}} + \eta \mathcal{L}_{\text{router}}, \quad (5)$$

where the auxiliary terms encourage marker sufficiency ($\mathcal{L}_{\text{suf}}$), marker diversity ($\mathcal{L}_{\text{div}}$), marker compression/sparsity ($\mathcal{L}_{\text{cmp}}$), and stable router logits and balanced token-choice routing ($\mathcal{L}_{\text{router}}$). $\mathcal{L}_{\text{CE}}$ is the class-weighted prediction loss. The nonnegative coefficients α , β , γ , and η set the relative influence of the four auxiliary objectives. Increasing one coefficient emphasizes its behavior but may reduce attention to the primary prediction objective, so any selection of these coefficients must use validation data rather than test performance. They should not be interpreted as independent performance gains. Sufficiency discourages selecting markers that lose predictive signal; diversity discourages redundant selections; compression encourages a compact set; and the router term stabilizes token-choice utilization.

In expert-choice routing, each recursion step selects a fixed number of the highest-scoring tokens. This fixed capacity directly controls how many tokens are processed at each step, so an additional load-balancing loss is not needed (Bae et al. 2025). To quantify computational cost, we report FLOPs,

a hardware-independent count of the arithmetic operations performed by the Transformer stack:

$$\Phi_{\text{eff}} = \sum_{t=1}^K (4a_t^2 d + 4a_t d d_{\text{ff}}), \quad (6)$$

where a_t is the mean number of active tokens at recursion step t , d is the model width, and d_{ff} is the hidden dimension of the feed-forward network. The first term grows quadratically with the active-token count and represents attention-like token interactions, while the second term represents feed-forward work and grows linearly with a_t . Φ_{eff} sums these costs over the executed recursion steps. Routing reduces computation when $a_t < M$ at later steps; because the same f_θ is reused, this compute saving is separate from the parameter saving obtained through weight sharing. The practical trade-off is controlled by K and the routing capacity: more active tokens or deeper recursion can improve refinement, but increases FLOPs and may provide diminishing returns.

Appendix E: Extended Results

Model	Type	N_R	Param	FLOPs	Single-cell macro-F1 $\uparrow$								Multi-omics macro-F1 $\uparrow$					Avg
					Segerst.	Lung	Oesoph.	Baron	Muraro	T-cell	Spleen	Xin	BLCA	STAD	PAN-2M	PAN-3M		
Vanilla	–	4	300K	1.00 $\times$	62.5 $\pm$ 11.2	72.5 $\pm$ 2.5	58.7 $\pm$ 4.2	61.0 $\pm$ 8.5	80.2 $\pm$ 4.3	52.1 $\pm$ 3.5	48.3 $\pm$ 3.7	74.0 $\pm$ 12.1	40.6 $\pm$ 5.8	44.5 $\pm$ 7.5	75.6 $\pm$ 3.5	95.2 $\pm$ 1.5	66.1 $\pm$ 5.3	
Recursive	–	2	75K	0.50 $\times$	65.8 $\pm$ 4.4	71.7 $\pm$ 2.8	57.3 $\pm$ 2.9	65.6 $\pm$ 2.8	79.8 $\pm$ 6.5	53.0 $\pm$ 2.0	48.8 $\pm$ 2.3	68.2 $\pm$ 2.4	41.4 $\pm$ 1.3	46.0 $\pm$ 3.4	75.7 $\pm$ 2.9	94.9 $\pm$ 1.5	66.3 $\pm$ 3.5	
	–	3	75K	0.75 $\times$	64.5 $\pm$ 3.1	72.7 $\pm$ 2.0	57.5 $\pm$ 2.1	67.1 $\pm$ 8.7	78.7 $\pm$ 4.9	49.8 $\pm$ 4.4	49.4 $\pm$ 3.3	67.8 $\pm$ 7.3	43.2 $\pm$ 6.2	45.7 $\pm$ 3.7	74.6 $\pm$ 2.7	95.0 $\pm$ 1.4	66.0 $\pm$ 4.1	
	–	4	75K	1.00 $\times$	61.4 $\pm$ 7.3	72.9 $\pm$ 2.9	56.9 $\pm$ 4.3	65.9 $\pm$ 2.7	75.8 $\pm$ 4.1	50.5 $\pm$ 3.5	49.4 $\pm$ 3.4	72.2 $\pm$ 5.2	40.2 $\pm$ 1.1	43.2 $\pm$ 6.0	72.3 $\pm$ 1.8	94.3 $\pm$ 1.1	65.5 $\pm$ 3.7	
MoR	Expert	2	75K	0.42 $\times$	65.0 $\pm$ 6.2	72.9 $\pm$ 1.7	57.0 $\pm$ 3.5	58.0 $\pm$ 6.1	72.3 $\pm$ 2.9	50.6 $\pm$ 4.9	51.7 $\pm$ 1.4	68.0 $\pm$ 8.3	39.7 $\pm$ 12.3	49.7 $\pm$ 6.3	77.7 $\pm$ 1.4	94.6 $\pm$ 1.3	65.9 $\pm$ 4.7	
	Expert	3	75K	0.59 $\times$	60.7 $\pm$ 6.9	74.2 $\pm$ 1.7	58.4 $\pm$ 3.6	61.6 $\pm$ 2.3	69.3 $\pm$ 7.2	52.3 $\pm$ 2.4	47.5 $\pm$ 3.0	65.7 $\pm$ 7.5	38.0 $\pm$ 8.0	42.3 $\pm$ 4.4	77.0 $\pm$ 1.1	94.6 $\pm$ 1.6	64.5 $\pm$ 4.3	
	Expert	4	75K	0.76 $\times$	64.3 $\pm$ 11.6	72.3 $\pm$ 2.7	55.3 $\pm$ 2.6	68.9 $\pm$ 3.7	74.8 $\pm$ 4.7	50.9 $\pm$ 4.7	53.2 $\pm$ 1.3	66.1 $\pm$ 9.7	43.7 $\pm$ 2.8	45.2 $\pm$ 4.1	75.5 $\pm$ 3.0	94.3 $\pm$ 1.3	65.8 $\pm$ 4.4	
	Token	2	75K	0.42 $\times$	57.4 $\pm$ 6.6	72.2 $\pm$ 2.7	57.7 $\pm$ 1.8	66.2 $\pm$ 8.5	82.5 $\pm$ 7.1	50.7 $\pm$ 2.2	50.6 $\pm$ 2.5	68.9 $\pm$ 6.0	35.7 $\pm$ 7.8	47.7 $\pm$ 4.0	76.4 $\pm$ 4.6	94.3 $\pm$ 2.8	66.6*	
	Token	3	75K	0.52 $\times$	55.1 $\pm$ 5.0	70.5 $\pm$ 3.3	56.6 $\pm$ 3.4	64.0 $\pm$ 4.6	75.9 $\pm$ 4.7	55.8 $\pm$ 3.0	49.2 $\pm$ 2.1	70.1 $\pm$ 3.5	41.3 $\pm$ 8.6	46.5 $\pm$ 4.5	77.2 $\pm$ 3.9	93.6 $\pm$ 2.1	65.6*	
	Token	4	75K	0.56 $\times$	60.7 $\pm$ 5.8	73.3 $\pm$ 2.4	59.8 $\pm$ 3.7	65.2 $\pm$ 5.5	81.6 $\pm$ 2.7	50.4 $\pm$ 2.2	51.1 $\pm$ 3.1	62.3 $\pm$ 5.7	40.6 $\pm$ 6.9	48.8 $\pm$ 3.6	78.0 $\pm$ 4.6	95.3 $\pm$ 1.6	67.0 $\pm$ 3.6	
bioMoR (ours)	Expert	2	75K	0.42 $\times$	72.4 $\pm$ 2.9	80.3$\pm$1.5	71.0 $\pm$ 1.5	85.2$\pm$2.9	83.8 $\pm$ 3.1	69.2 $\pm$ 0.7	60.3 $\pm$ 1.9	76.3 $\pm$ 9.6	28.2 $\pm$ 4.5	43.6 $\pm$ 3.9	85.5 $\pm$ 1.9	96.0 $\pm$ 1.3	71.4 $\pm$ 2.8	
	Expert	3	75K	0.59 $\times$	76.1$\pm$2.9	80.1 $\pm$ 0.3	69.2 $\pm$ 1.6	81.6 $\pm$ 2.7	85.1$\pm$7.6	69.6 $\pm$ 0.6	61.4$\pm$1.7	80.1$\pm$7.0	34.2 $\pm$ 6.5	49.2 $\pm$ 5.0	86.0 $\pm$ 1.0	96.3 $\pm$ 0.8	72.7 $\pm$ 3.1	
	Expert	4	75K	0.76 $\times$	72.2 $\pm$ 4.5	80.0 $\pm$ 1.2	70.8 $\pm$ 1.6	81.9 $\pm$ 1.9	83.7 $\pm$ 4.9	70.0$\pm$1.9	59.7 $\pm$ 0.8	72.7 $\pm$ 5.5	47.1$\pm$7.1	46.0 $\pm$ 2.6	85.8 $\pm$ 1.5	98.0$\pm$0.4	72.6 $\pm$ 2.8	
+ Token choice	Token	2	75K	0.42 $\times$	74.4 $\pm$ 6.8	79.4 $\pm$ 1.3	71.1 $\pm$ 2.4	82.9 $\pm$ 5.0	83.8 $\pm$ 3.2	68.6 $\pm$ 1.7	60.3 $\pm$ 1.2	69.1 $\pm$ 6.6	45.0 $\pm$ 13.8	44.2 $\pm$ 3.5	85.1 $\pm$ 1.5	95.5 $\pm$ 2.2	71.6 $\pm$ 1.6	
	Token	3	75K	0.52 $\times$	73.6 $\pm$ 7.0	80.2 $\pm$ 1.5	71.0 $\pm$ 1.2	83.9 $\pm$ 3.0	78.4 $\pm$ 4.8	68.6 $\pm$ 0.9	61.0 $\pm$ 1.0	77.2 $\pm$ 5.3	39.6 $\pm$ 13.7	45.4 $\pm$ 4.8	84.8 $\pm$ 2.3	95.8 $\pm$ 1.0	71.6 $\pm$ 1.5	
	Token	4	75K	0.56 $\times$	75.8 $\pm$ 2.8	79.7 $\pm$ 0.9	70.8 $\pm$ 2.1	82.9 $\pm$ 4.0	81.9 $\pm$ 6.8	69.8 $\pm$ 1.2	61.0 $\pm$ 0.6	76.1 $\pm$ 8.9	39.5 $\pm$ 5.9	51.4$\pm$5.0	86.0$\pm$1.9	96.4 $\pm$ 1.6	73.0$\pm$3.0	

Table 3: **Complete architecture ladder (5-fold CV).** Macro-F1 (mean $\pm$ SD) for all 11 classification datasets. Parameters and Transformer-stack FLOPs are reported relative to the four-layer Vanilla Transformer. PM, PC, and 3M denote pan-cancer mutation+CNV, expression, and tri-modal inputs, respectively. This expands the representative eight-dataset table in the main paper; bold, underlined values mark the best result in each dataset column.

Model	Type	N_R	Param	FLOPs	Single-cell balanced acc. $\uparrow$								Multi-omics balanced acc. $\uparrow$				Avg
					Segerst.	Lung	Oesoph.	Baron	Muraro	T-cell	Spleen	Xin	BLCA	STAD	PAN-2M	PAN-3M	
Vanilla	–	4	300K	1.00×	71.7 $\pm$ 10.3	75.2 $\pm$ 1.5	60.0 $\pm$ 2.9	67.7 $\pm$ 2.6	79.1 $\pm$ 3.3	55.2 $\pm$ 5.7	55.3 $\pm$ 1.4	73.6 $\pm$ 8.7	48.3 $\pm$ 2.8	48.4 $\pm$ 3.8	74.1 $\pm$ 4.0	94.9 $\pm$ 1.9	67.1*
Recursive	–	2	75K	0.50×	69.2 $\pm$ 11.0	75.9 $\pm$ 1.5	61.1 $\pm$ 4.1	71.0 $\pm$ 5.6	77.1 $\pm$ 4.6	57.4 $\pm$ 6.1	57.8 $\pm$ 1.5	73.1 $\pm$ 9.7	48.9 $\pm$ 5.4	49.3 $\pm$ 3.9	72.7 $\pm$ 4.2	94.7 $\pm$ 2.1	67.1*
	–	3	75K	0.75×	74.9 $\pm$ 9.7	75.0 $\pm$ 1.7	61.1 $\pm$ 3.2	71.1 $\pm$ 3.4	79.8 $\pm$ 5.4	55.7 $\pm$ 4.7	57.9 $\pm$ 1.8	74.1 $\pm$ 9.8	46.0 $\pm$ 3.7	50.2 $\pm$ 2.5	71.7 $\pm$ 5.2	95.5 $\pm$ 2.8	67.5*
	–	4	75K	1.00×	69.6 $\pm$ 9.1	75.3 $\pm$ 1.9	60.2 $\pm$ 3.6	70.8 $\pm$ 6.9	77.8 $\pm$ 2.4	56.5 $\pm$ 4.6	56.9 $\pm$ 1.5	76.5 $\pm$ 9.4	48.5 $\pm$ 2.8	49.6 $\pm$ 3.5	74.4 $\pm$ 1.6	93.6 $\pm$ 1.7	67.2*
MoR	Expert	2	75K	0.42×	65.9 $\pm$ 3.6	75.6 $\pm$ 1.3	61.2 $\pm$ 3.0	68.7 $\pm$ 3.8	76.7 $\pm$ 5.0	58.1 $\pm$ 2.3	57.8 $\pm$ 1.3	75.2 $\pm$ 5.8	46.5 $\pm$ 3.0	51.4 $\pm$ 4.1	76.4 $\pm$ 3.1	94.8 $\pm$ 1.5	67.7*
	Expert	3	75K	0.59×	70.9 $\pm$ 4.3	76.2 $\pm$ 1.4	61.4 $\pm$ 3.2	68.4 $\pm$ 2.0	81.4 $\pm$ 5.8	56.0 $\pm$ 5.5	58.0 $\pm$ 1.5	75.3 $\pm$ 6.7	47.1 $\pm$ 4.5	49.1 $\pm$ 2.6	74.6 $\pm$ 5.3	92.5 $\pm$ 3.1	67.3*
	Expert	4	75K	0.76×	69.2 $\pm$ 6.1	76.5 $\pm$ 1.8	62.7 $\pm$ 3.7	70.9 $\pm$ 2.7	75.5 $\pm$ 2.3	56.6 $\pm$ 6.1	57.8 $\pm$ 2.5	74.1 $\pm$ 10.0	49.8 $\pm$ 5.0	47.0 $\pm$ 1.3	74.6 $\pm$ 3.1	93.7 $\pm$ 3.1	67.8*
	Token	2	75K	0.42×	62.6 $\pm$ 5.1	76.6 $\pm$ 1.9	61.5 $\pm$ 2.1	65.2 $\pm$ 1.8	74.4 $\pm$ 6.7	57.6 $\pm$ 4.1	55.6 $\pm$ 1.5	65.5 $\pm$ 9.3	45.2 $\pm$ 2.8	52.1 $\pm$ 5.8	77.7 $\pm$ 2.2	93.3 $\pm$ 1.9	66.4*
	Token	3	75K	0.52×	67.3 $\pm$ 11.7	77.8 $\pm$ 1.5	63.9 $\pm$ 2.6	69.9 $\pm$ 6.5	77.5 $\pm$ 5.8	52.4 $\pm$ 5.2	56.5 $\pm$ 2.3	68.6 $\pm$ 9.8	47.3 $\pm$ 3.1	49.9 $\pm$ 4.7	74.5 $\pm$ 6.0	94.6 $\pm$ 1.9	67.2*
	Token	4	75K	0.56×	66.4 $\pm$ 8.0	76.1 $\pm$ 2.2	61.4 $\pm$ 2.3	67.3 $\pm$ 4.5	80.6 $\pm$ 8.5	56.6 $\pm$ 4.1	56.5 $\pm$ 1.4	72.9 $\pm$ 10.9	46.7 $\pm$ 7.8	47.4 $\pm$ 3.3	73.5 $\pm$ 2.2	93.7 $\pm$ 4.6	66.7*
bioMoR (ours)	Expert	2	75K	0.42×	73.1 $\pm$ 3.6	81.5 $\pm$ 1.0	72.0 $\pm$ 1.7	82.2 $\pm$ 3.1	84.9 $\pm$ 6.2	70.2 $\pm$ 1.5	65.4$\pm$1.5	75.0 $\pm$ 8.3	48.9 $\pm$ 3.3	56.1$\pm$6.3	83.4 $\pm$ 1.1	96.4$\pm$1.2	73.3*
	Expert	3	75K	0.59×	77.5 $\pm$ 4.8	80.7 $\pm$ 1.6	71.2 $\pm$ 0.3	87.4$\pm$4.8	82.5 $\pm$ 8.7	70.8 $\pm$ 1.5	64.3 $\pm$ 0.9	78.3 $\pm$ 6.7	49.0 $\pm$ 0.8	50.2 $\pm$ 1.6	85.5 $\pm$ 1.5	95.6 $\pm$ 2.5	73.5*
	Expert	4	75K	0.76×	78.8 $\pm$ 4.9	80.4 $\pm$ 0.6	72.1 $\pm$ 1.6	85.8 $\pm$ 2.7	88.9$\pm$4.0	70.4 $\pm$ 1.6	64.7 $\pm$ 1.6	79.7 $\pm$ 5.2	48.7 $\pm$ 7.5	51.0 $\pm$ 4.0	82.9 $\pm$ 3.3	95.4 $\pm$ 1.3	73.9*
+ Token choice	Token	4	75K	0.56×	73.3 $\pm$ 7.0	82.0$\pm$1.5	73.2$\pm$1.6	86.4 $\pm$ 3.8	83.4 $\pm$ 3.7	71.3$\pm$1.4	64.1 $\pm$ 1.1	81.7 $\pm$ 5.8	48.3 $\pm$ 2.2	46.7 $\pm$ 3.0	83.4 $\pm$ 2.6	96.1 $\pm$ 1.9	73.3*

Table 4: **Balanced accuracy across the classification suite (5-fold CV).** Each populated cell reports mean $\pm$ SD balanced accuracy over the same five folds and architecture ladder used for macro-F1. Bold, underlined values mark the best result in each dataset column; starred averages exclude unavailable runs from the corresponding architecture configuration.

Appendix F: Statistical Significance

We use the paired Wilcoxon signed-rank test introduced by [Wilcoxon \(1945\)](#) and follow the multiple-dataset classifier-comparison guidance of [Demšar \(2006\)](#).

Table 5: **Per-dataset macro-F1 (5-fold CV) and bioMoR’s improvement over the strongest baseline.** Each column is one dataset; bold marks the best model. Δ vs best is bioMoR minus the per-dataset maximum of {Vanilla, Recursive, MoR}. bioMoR wins 11/12 (one-sided Wilcoxon signed-rank $p=0.0007$; vs. Recursive and vs. MoR individually 12/12, $p=0.0002$).

Model	Seger.	Lung	Oeso.	Baron	Muraro	T-cell	Spleen	Xin	BLCA	STAD	PAN-2M	PAN-3M
Vanilla	62.5	72.5	58.7	65.7	80.2	52.1	48.3	74.0	40.6	44.5	75.6	95.2
Recursive	61.4	72.9	56.9	70.3	75.8	50.5	49.4	72.2	40.2	43.2	72.3	94.3
MoR	64.3	72.3	55.3	68.9	74.8	50.9	53.2	66.1	43.7	45.2	75.5	94.3
bioMoR	72.2	80.0	70.8	81.9	83.7	70.0	59.7	72.7	47.1	46.0	85.8	98.0
Δ vs best	+7.9	+7.1	+12.1	+11.6	+3.5	+17.9	+6.5	-1.3	+3.4	+0.8	+10.2	+2.8
Wilcoxon p	$p = 0.0007$ (one-sided, bioMoR > best baseline, $n = 12$)											
Win	✓	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓

Table 6: **Per-dataset balanced accuracy (5-fold CV) and bioMoR’s improvement over the strongest baseline.** Layout as in Table 5. bioMoR wins 11/12 (one-sided Wilcoxon signed-rank $p=0.0007$; vs. Vanilla and vs. Recursive individually 12/12, $p=0.0002$).

Model	Seger.	Lung	Oeso.	Baron	Muraro	T-cell	Spleen	Xin	BLCA	STAD	PAN-2M	PAN-3M
Vanilla	71.7	75.2	60.0	67.7	79.1	55.2	55.3	73.6	48.3	48.4	74.1	94.9
Recursive	69.6	75.3	60.2	70.8	77.8	56.5	56.9	76.5	48.5	49.6	74.4	93.6
MoR	69.2	76.5	62.7	70.9	75.5	56.6	57.8	74.1	49.8	47.0	74.6	93.7
bioMoR	78.8	80.4	72.1	85.8	88.9	70.4	64.7	79.7	48.7	51.0	82.9	95.4
Δ vs best	+7.1	+3.9	+9.4	+15.0	+9.8	+13.8	+6.9	+3.2	-1.1	+1.4	+8.3	+0.5
Wilcoxon p	$p = 0.0007$ (one-sided, bioMoR > best baseline, $n = 12$)											
Win	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	✓	✓

Appendix G: Extended Survival Prediction

To test whether bioMoR transfers beyond classification, we evaluate survival-risk prediction on the PAN-2M pan-cancer cohort and four cancer-specific TCGA cohorts: KIRP, COAD, UCEC, and HNSC. The comparison includes bioMoR-Expert and bioMoR-Token together with the Vanilla, Recursive, MoR-Expert, and MoR-Token baselines. Performance is measured using Harrell’s C-index; a larger value indicates better concordance between the predicted risk ranking and observed survival outcomes.

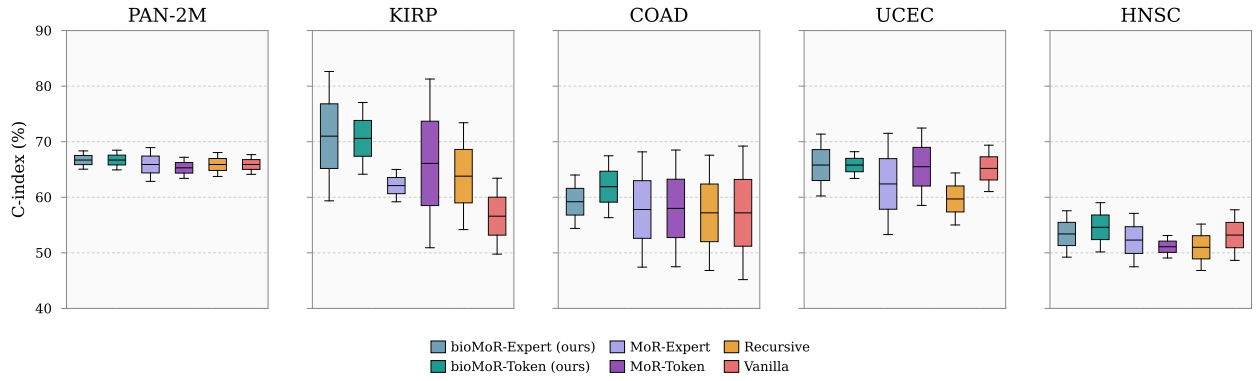


Figure 1: **Survival prediction across pan-cancer and TCGA cohorts.** Fold-level C-index distributions for PAN-2M, KIRP, COAD, UCEC, and HNSC. The bioMoR variants remain competitive across the pan-cancer and cancer-specific settings, supporting the transfer of biology-guided adaptive recursion to a time-to-event endpoint. Higher values indicate better concordance.

Appendix H: Embedding Visualization with UMAP

To examine whether predictive performance is reflected in the learned representation, we freeze each model’s penultimate embedding, fit a logistic-regression probe on the training samples, and evaluate it on held-out samples. Each dataset panel compares Vanilla, Recursive, MoR, and bioMoR under the same protocol.

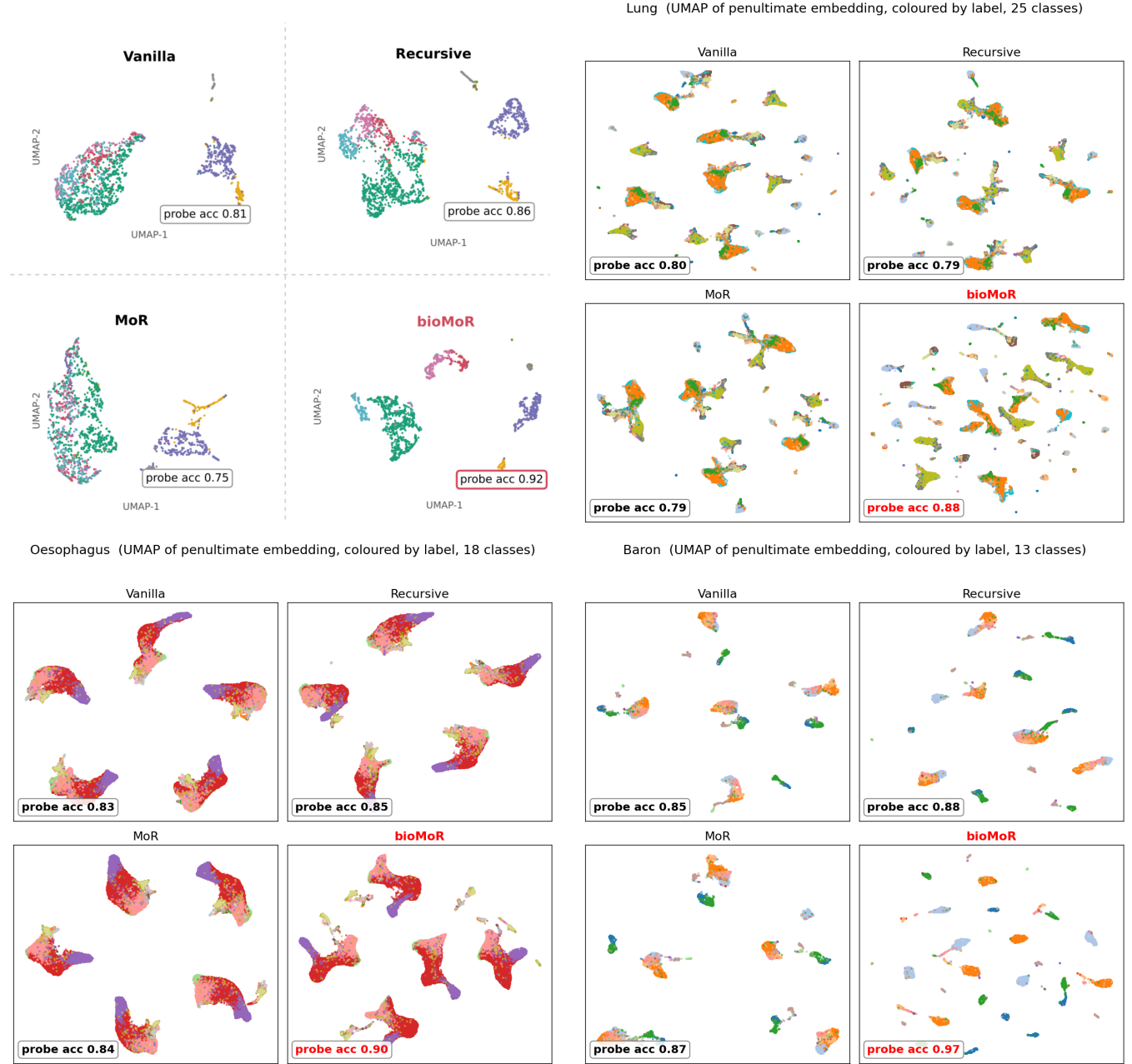


Figure 2: **Frozen-embedding UMAP comparisons for single-cell datasets (part 1).** Each dataset-level panel shows the four architecture variants, colors points by the ground-truth class, and reports held-out linear-probe accuracy within each subpanel. The Segerstolpe panel is the same figure used in the main paper.

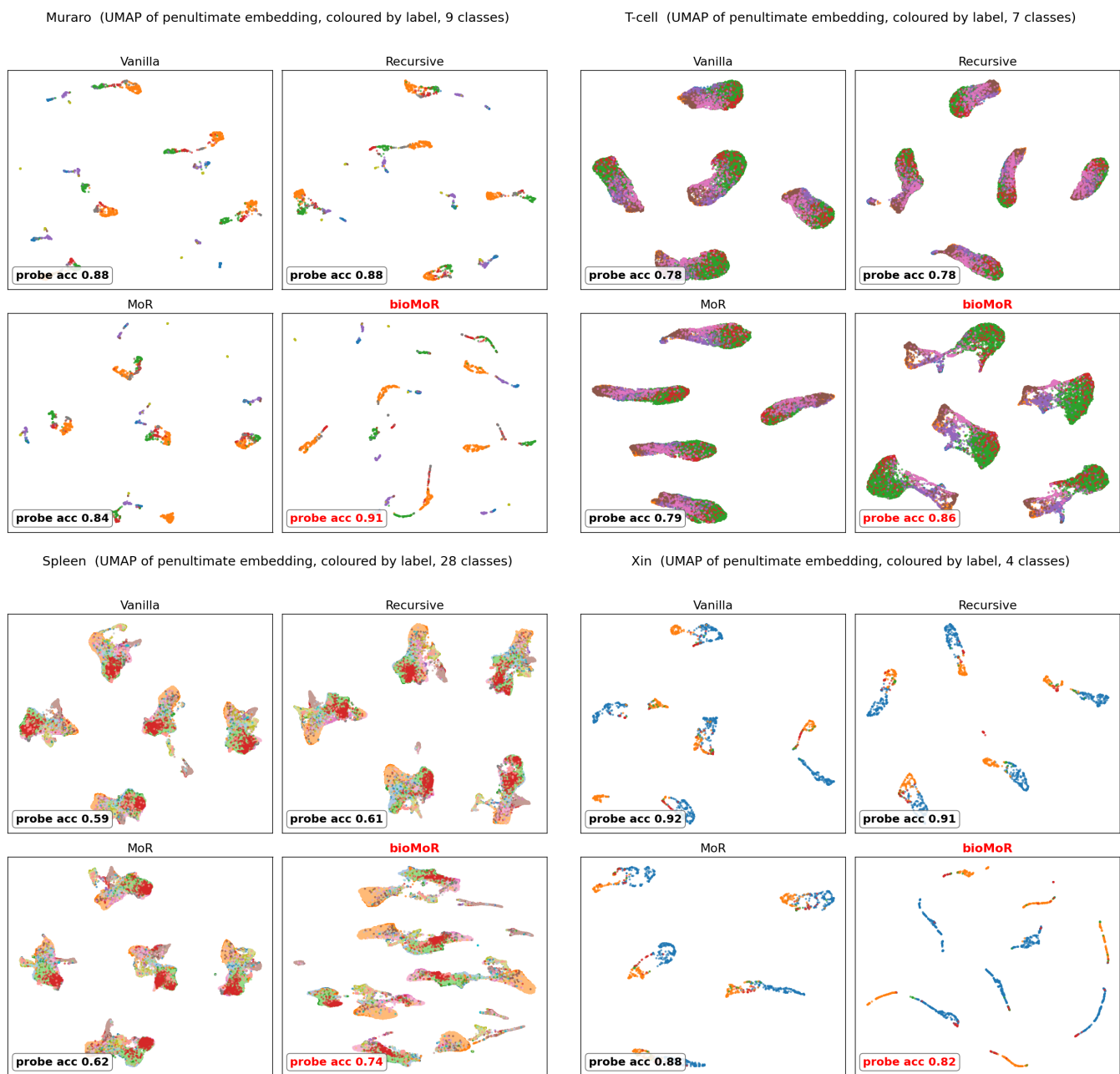
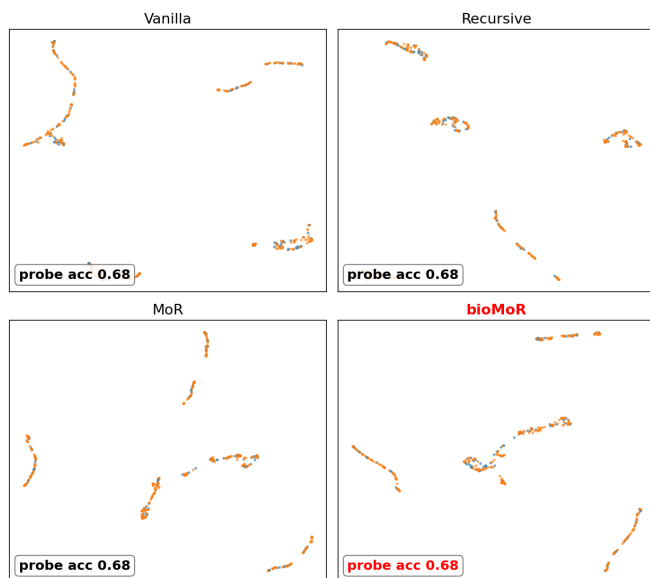
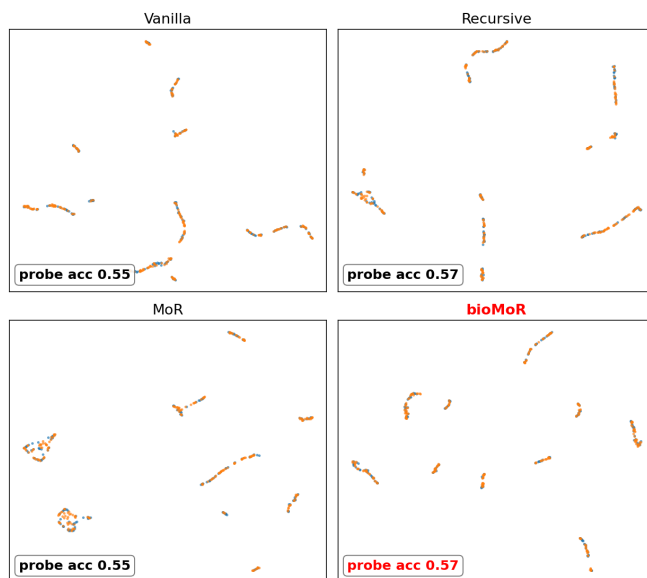


Figure 3: **Frozen-embedding UMAP comparisons for single-cell datasets (part 2).** The plotting, coloring, and held-out linear-probe protocol match Figure 2.

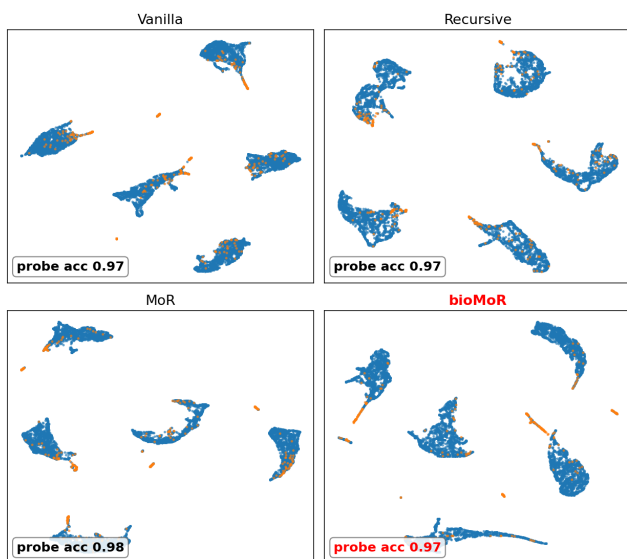
BLCA (UMAP of penultimate embedding, coloured by label, 2 classes)



STAD (stomach) (UMAP of penultimate embedding, coloured by label, 2 classes)



Pan-cancer (meta/pri) (UMAP of penultimate embedding, coloured by label, 2 classes)



Pan-cancer (tri-modal) (UMAP of penultimate embedding, coloured by label, 2 classes)

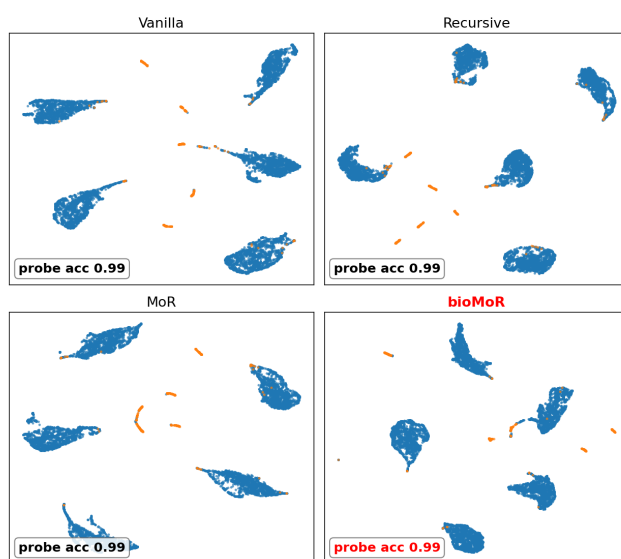


Figure 4: **Frozen-embedding UMAP comparisons for multi-omics datasets.** The plotting, coloring, and held-out linear-probe protocol match Figure 2; PAN-2M and PAN-3M denote the bi-modal and tri-modal pan-cancer cohorts, respectively.

Appendix I: Ablation Studies

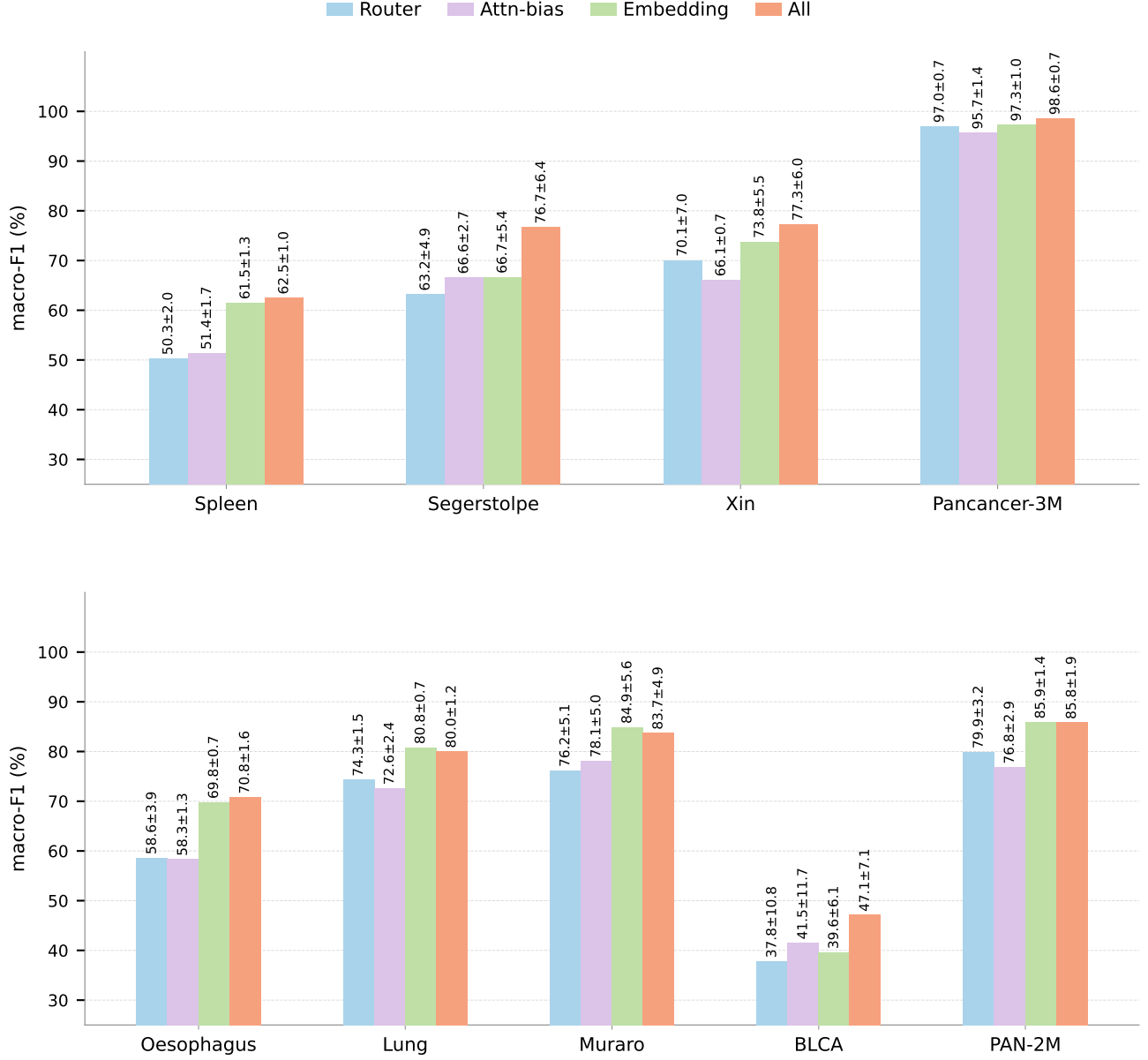


Figure 5: **Biological-injection ablation at recursion depth $K = 4$.** Macro-F1 (mean with fold-level variability) on the same four representative datasets shown in the main paper—Spleen, Segerstolpe, Xin, and Pancancer-3M—when biological information is applied only through the router, attention bias, or embedding, compared with the complete expert-choice bioMoR configuration. The first row matches the main-paper dataset order, labels, and values exactly; the second row extends the comparison to Oesophagus, Lung, Muraro, BLCA, and PAN-2M.

Appendix J: Training and Validation Dynamics

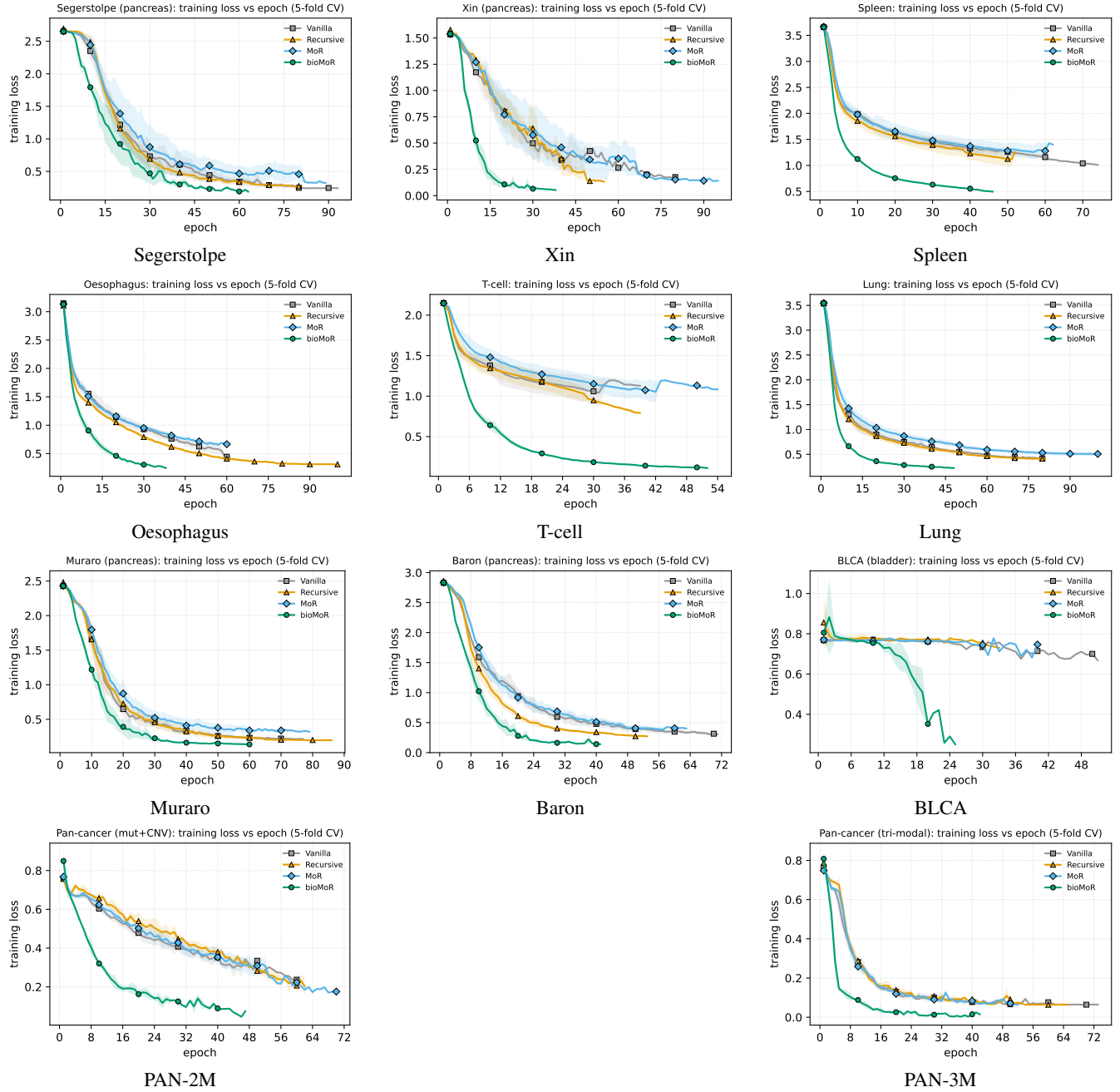


Figure 6: **Training loss across epochs.** Fold-aggregated training-loss trajectories for the available single-cell and multi-omics datasets used in the RQ3 convergence analysis.

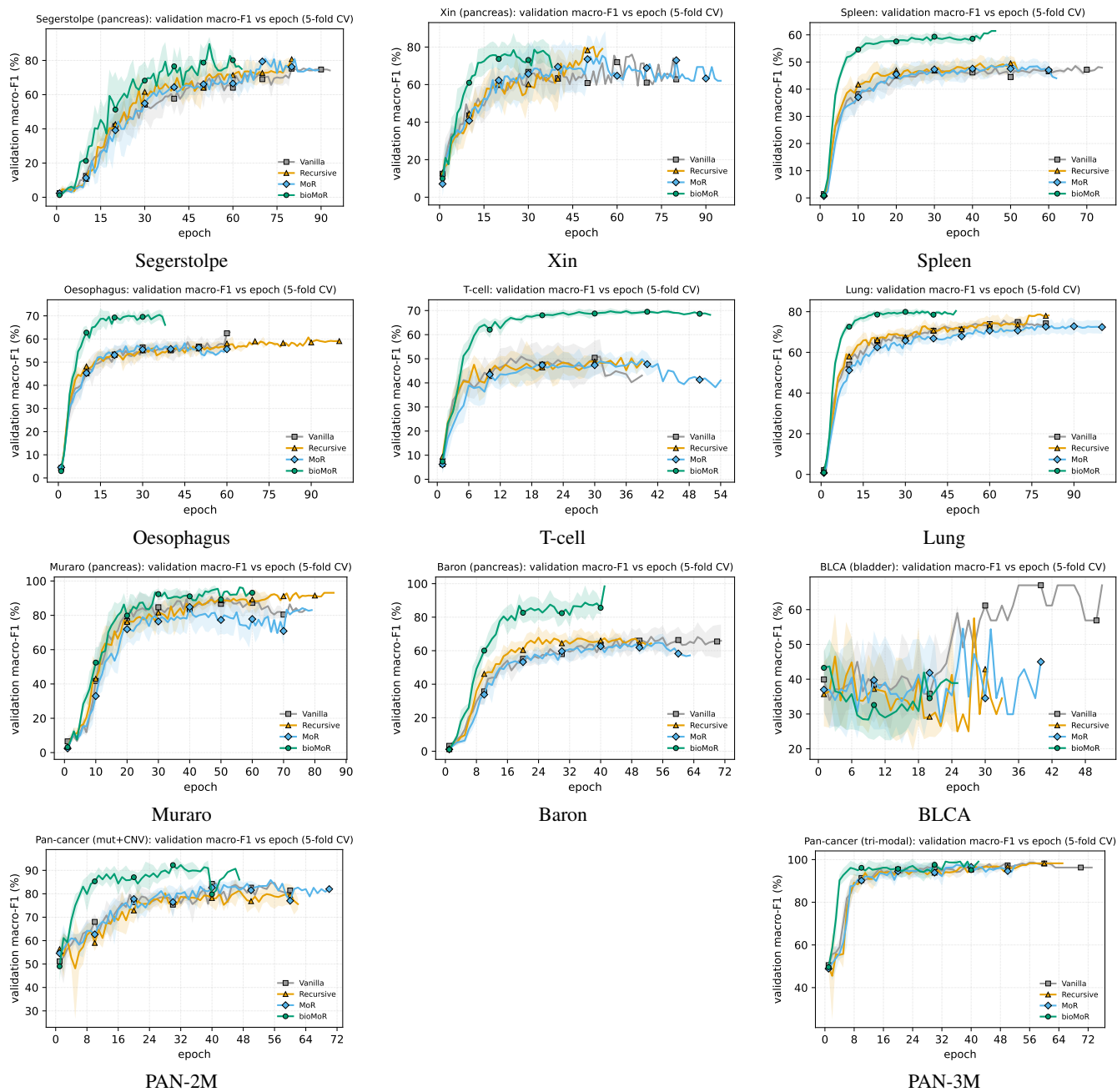


Figure 7: **Validation macro-F1 across epochs.** Fold-aggregated validation macro-F1 trajectories for all datasets with available RQ3 validation curves. Higher values indicate better validation performance.

Appendix K: Accuracy-Efficiency Trade-offs

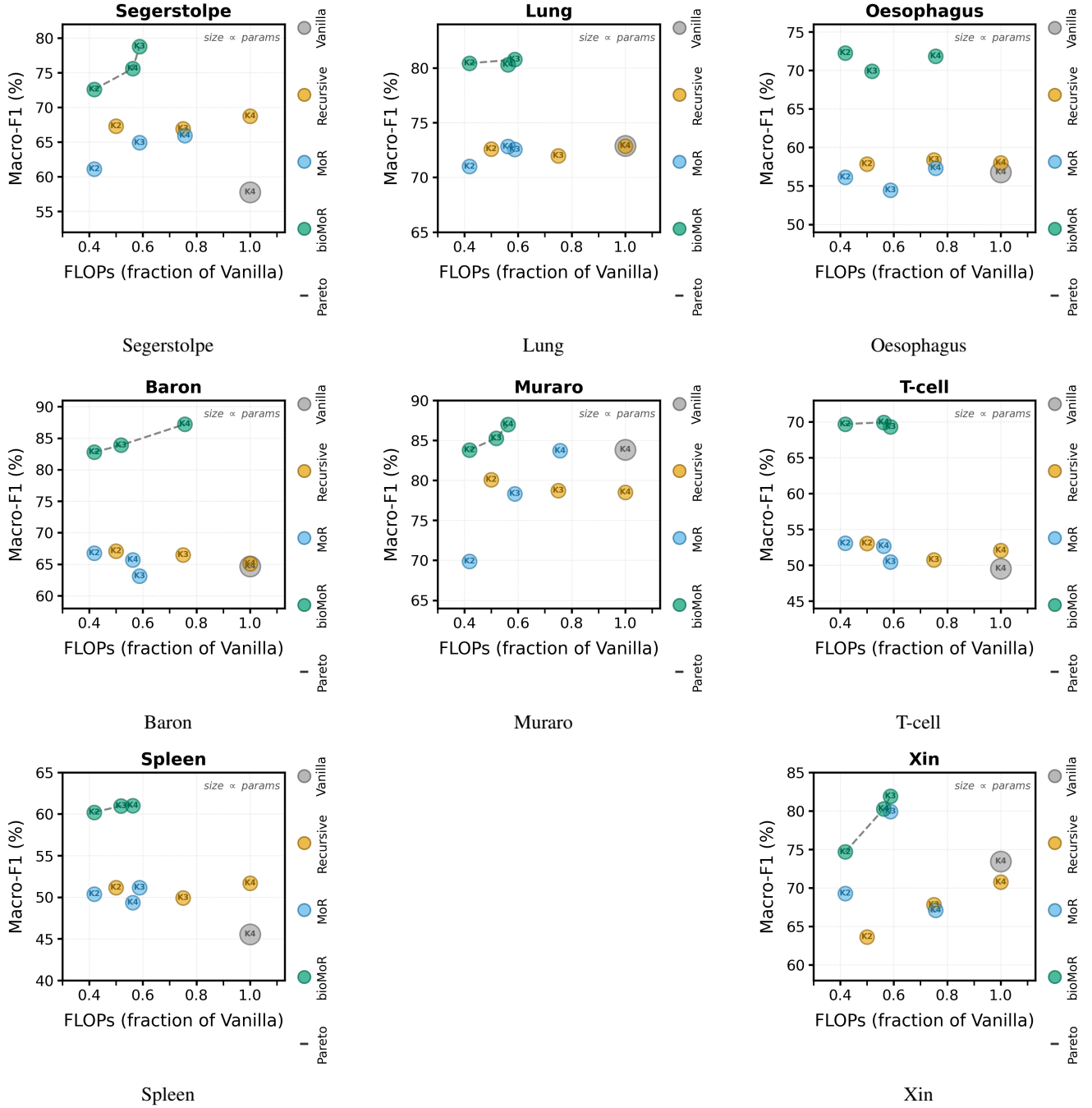


Figure 8: **Validation macro-F1 versus effective computation (part 1).** Dataset-level Pareto views compare the accuracy–efficiency trade-off across architectures and recursion settings; points toward the upper-left are preferred.

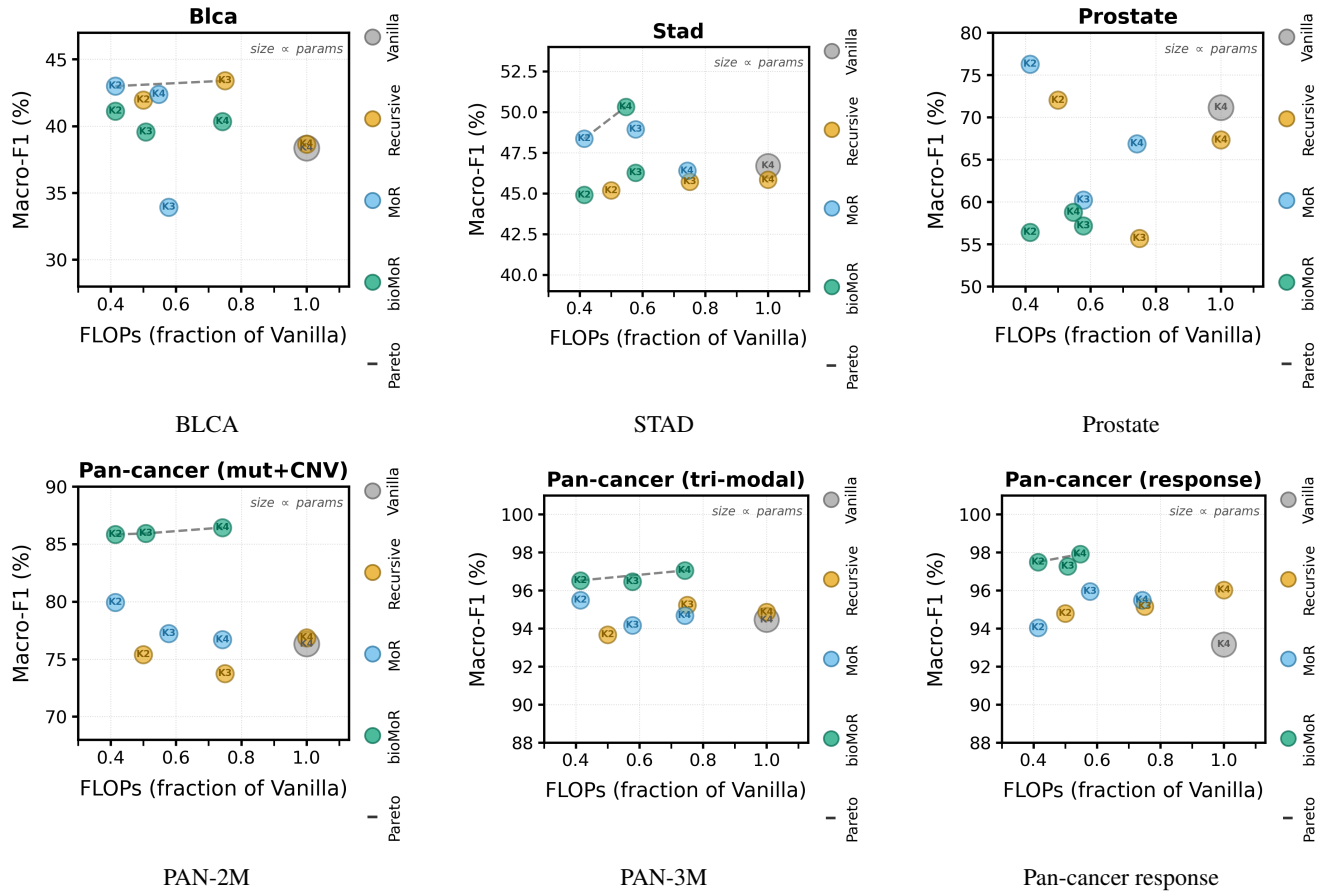


Figure 9: **Validation macro-F1 versus effective computation (part 2).** Multi-omics Pareto views use the same axes and interpretation as Figure 8.

Appendix L: Case Study: PAN-3M

Pathway Markers by Recursion Depth ($M=1257$, $K=4$, Macro-F1 96.9 ± 1.4)

Per-depth keep/erase rankings for the tri-modal pan-cancer cohort (PAN-3M; mutation+CNV+expression, 8586 samples, 2 classes) under canonical bioMoR at $K=4$ expert-choice routing with capacity floor $\text{RMT_CAP_FLOOR}=0.5625$ (schedule $c = (1.00, 0.75, 0.5625, 0.5625)$). The pathway view tokenises all 1257 Reactome pathways using the learned pathway graph. The expert-choice funnel keeps 56% of pathway tokens to full depth (707/1257): tokens drop at depth 2 (to 75%) and again at depth 3 (to 56%), then hold. Pathway-view Macro-F1 is 96.9. Mean recursion depth $d_m \in [1, 4]$ is averaged over the 5 held-out folds. For each depth $d \in \{2, 3, 4\}$ we list up to 25 *kept* markers still active at depth d (rounded $d_m \geq d$, deepest first) and up to 25 *erased* markers that exit entering depth d (rounded $d_m = d-1$). Depth 1 keeps all M tokens.

Depth 2.

Kept — active at depth 2

#	Pathway	d_m
1	Intra-Golgi and retrograde Golgi-to-ER traffic	3.97
2	Glycerophospholipid biosynthesis	3.96
3	Post-translational modification: synthesis of GPI-anchored proteins	3.95
4	Golgi-to-ER retrograde transport	3.95
5	Factors involved in megakaryocyte development and platelet production	3.95
6	Semaphorin interactions	3.95
7	RUNX1 regulates genes involved in megakaryocyte differentiation and p...	3.95
8	FOXO-mediated transcription	3.94
9	Phospholipid metabolism	3.94
10	Kinesins	3.94
11	Chaperonin-mediated protein folding	3.93
12	Deubiquitination	3.92
13	Class I MHC mediated antigen processing & presentation	3.92
14	Signaling by Interleukins	3.92
15	SLC-mediated transmembrane transport	3.90
16	Coagulation pathway	3.90
17	Synthesis of PA	3.90
18	Regulation of clotting cascade	3.90
19	Regulation of MTF-M-dependent genes involved in apoptosis	3.90
20	MHC class II antigen presentation	3.90
21	Cell surface interactions at the vascular wall	3.89
22	Cellular Senescence	3.88
23	NOTCH3 Intracellular Domain Regulates Transcription	3.88
24	Ca2+ pathway	3.87
25	Mitotic Prometaphase	3.85

Erased — exit entering depth 2

#	Pathway	d_m
1	GABA synthesis, release, reuptake and degradation	1.48
2	APC/C:Cdh1 mediated degradation of Cdc20 and other APC/C:Cdh1 targets...	1.47
3	Sensory processing of sound by inner hair cells of the cochlea	1.47
4	Unblocking of NMDA receptors, glutamate binding and activation	1.47
5	MAP3K8 (TPL2)-dependent MAPK1/3 activation	1.45
6	SHC-mediated cascade:FGFR3	1.45
7	Phosphorylated BMAL1:CLOCK (ARNTL:CLOCK) activates expression of core...	1.43
8	Degradation of CRY and PER proteins	1.43
9	Transcription of E2F targets under negative control by p107 (RBL1) an...	1.41
10	Activation of G protein gated Potassium channels	1.41
11	Fibronectin matrix formation	1.40
12	MyD88-independent TLR4 cascade	1.39
13	Disorders of Developmental Biology	1.36
14	Trafficking of AMPA receptors	1.34
15	E3 ubiquitin ligases ubiquitinate target proteins	1.28
16	Anchoring fibril formation	1.22
17	JNK (c-Jun kinases) phosphorylation and activation mediated by activ...	1.21
18	FRS-mediated FGFR3 signaling	1.17
19	Paracetamol ADME	1.14
20	Aspirin ADME	1.10
21	Drug ADME	1.06

Depth 3.

Kept — active at depth 3		
#	Pathway	d_m
1	Intra-Golgi and retrograde Golgi-to-ER traffic	3.97
2	Glycerophospholipid biosynthesis	3.96
3	Post-translational modification: synthesis of GPI-anchored proteins	3.95
4	Golgi-to-ER retrograde transport	3.95
5	Factors involved in megakaryocyte development and platelet production	3.95
6	Semaphorin interactions	3.95
7	RUNX1 regulates genes involved in megakaryocyte differentiation and p . . .	3.95
8	FOXO-mediated transcription	3.94
9	Phospholipid metabolism	3.94
10	Kinesins	3.94
11	Chaperonin-mediated protein folding	3.93
12	Deubiquitination	3.92
13	Class I MHC mediated antigen processing & presentation	3.92
14	Signaling by Interleukins	3.92
15	SLC-mediated transmembrane transport	3.90
16	Coagulation pathway	3.90
17	Synthesis of PA	3.90
18	Regulation of clotting cascade	3.90
19	Regulation of MITF-M-dependent genes involved in apoptosis	3.90
20	MHC class II antigen presentation	3.90
21	Cell surface interactions at the vascular wall	3.89
22	Cellular Senescence	3.88
23	NOTCH3 Intracellular Domain Regulates Transcription	3.88
24	Ca2+ pathway	3.87
25	Mitotic Prometaphase	3.85

Erased — exit entering depth 3

#	Pathway	d_m
1	Formation of WDR5-containing histone-modifying complexes	2.50
2	COPII-mediated vesicle transport	2.50
3	TP53 Regulates Transcription of Genes Involved in Cytochrome C Release	2.49
4	HIV Life Cycle	2.49
5	Translation of Structural Proteins_9694635	2.49
6	Netrin-1 signaling	2.49
7	Glucagon-type ligand receptors	2.49
8	Signaling by BMP	2.49
9	Notch-HLH transcription pathway	2.49
10	Impaired BRCA2 binding to PALB2	2.49
11	Processive synthesis on the C-strand of the telomere	2.49
12	Nicotinate metabolism	2.49
13	Negative regulators of DDX58/IFIH1 signaling	2.49
14	Aberrant regulation of mitotic exit in cancer due to RB1 defects	2.48
15	Infection with Enterobacteria	2.48
16	ADORA2B mediated anti-inflammatory cytokines production	2.48
17	ABC-family protein mediated transport	2.48
18	Degradation of GLI1 by the proteasome	2.48
19	Striated Muscle Contraction	2.47
20	Proteasome assembly	2.47
21	Assembly of collagen fibrils and other multimeric structures	2.47
22	Sensory perception of sweet, bitter, and umami (glutamate) taste	2.47
23	Interleukin-10 signaling	2.47
24	HIV Transcription Initiation	2.47
25	Pausing and recovery of Tat-mediated HIV elongation	2.46

Depth 4.

Kept — active at depth 4		
#	Pathway	d_m
1	Intra-Golgi and retrograde Golgi-to-ER traffic	3.97
2	Glycerophospholipid biosynthesis	3.96
3	Post-translational modification: synthesis of GPI-anchored proteins	3.95
4	Golgi-to-ER retrograde transport	3.95
5	Factors involved in megakaryocyte development and platelet production	3.95
6	Semaphorin interactions	3.95
7	RUNX1 regulates genes involved in megakaryocyte differentiation and p...	3.95
8	FOXO-mediated transcription	3.94
9	Phospholipid metabolism	3.94
10	Kinesins	3.94
11	Chaperonin-mediated protein folding	3.93
12	Deubiquitination	3.92
13	Class I MHC mediated antigen processing & presentation	3.92
14	Signaling by Interleukins	3.92
15	SLC-mediated transmembrane transport	3.90
16	Coagulation pathway	3.90
17	Synthesis of PA	3.90
18	Regulation of clotting cascade	3.90
19	Regulation of MITF-M-dependent genes involved in apoptosis	3.90
20	MHC class II antigen presentation	3.90
21	Cell surface interactions at the vascular wall	3.89
22	Cellular Senescence	3.88
23	NOTCH3 Intracellular Domain Regulates Transcription	3.88
24	Ca2+ pathway	3.87
25	Mitotic Prometaphase	3.85

Erased — exit entering depth 4

#	Pathway	d_m
1	Signaling by MET	3.50
2	Synthesis of substrates in N-glycan biosynthesis	3.50
3	Signaling by NTRK2 (TRKB)	3.49
4	Tie2 Signaling	3.49
5	Depolymerization of the Nuclear Lamina	3.49
6	Complement cascade	3.49
7	COPI-mediated anterograde transport	3.49
8	HSF1-dependent transactivation	3.49
9	Differentiation of T cells	3.49
10	Amyloid fiber formation	3.49
11	Nuclear Events (kinase and transcription factor activation)	3.49
12	Respiratory Syncytial Virus Infection Pathway	3.48
13	Signaling by SCF-KIT	3.48
14	Translation of Replicase and Assembly of the Replication Transcriptio...	3.48
15	FLT3 Signaling	3.48
16	Regulation of MITF-M-dependent genes involved in lysosome biogenesis...	3.48
17	Mitochondrial tRNA aminoacylation	3.48
18	Triglyceride metabolism	3.48
19	Generation of second messenger molecules	3.48
20	Transcriptional regulation of granulopoiesis	3.48
21	CD28 dependent PI3K/Akt signaling	3.48
22	RNA Polymerase III Transcription Initiation	3.48
23	Metabolism of carbohydrates and carbohydrate derivatives	3.47
24	Heparan sulfate/heparin (HS-GAG) metabolism	3.47
25	Mitotic G2-G2/M phases	3.47

References

- Arik, S. O., and Pfister, T. 2021. TabNet: Attentive interpretable tabular learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 6679–6687.
- Bae, S.; Kim, Y.; Bayat, R.; Kim, S.; Ha, J.; et al. 2025. Mixture-of-recursions: Learning dynamic recursive depths for adaptive token-level computation. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Breiman, L. 2001. Random forests. *Machine Learning* 45:5–32.
- Chen, J.; Xu, H.; Tao, W.; Chen, Z.; Zhao, Y.; and Han, J.-D. J. 2023. Transformer for one stop interpretable cell type annotation. *Nature Communications* 14:223.
- Cui, H.; Wang, C.; Maan, H.; Pang, K.; Luo, F.; Duan, N.; and Wang, B. 2024. scgpt: Toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* 21:1470–1480.
- Dehghani, M.; Gouws, S.; Vinyals, O.; Uszkoreit, J.; and Kaiser, L. 2019. Universal transformers. In *International Conference on Learning Representations (ICLR)*.
- Elmarakeby, H. A.; Hwang, J.; Arafeh, R.; et al. 2021. Biologically informed deep neural network for prostate cancer discovery. *Nature* 598:348–352.
- Fu, Y.; Xu, J.; Tang, Z.; et al. 2020. A gene prioritization method based on a swine multi-omics knowledgebase and a deep learning model. *Communications Biology* 3:502.
- Gillespie, M.; Jassal, B.; Stephan, R.; Milacic, M.; Rothfels, K.; Senff-Ribeiro, A.; Griss, J.; et al. 2022. The reactome pathway knowledgebase 2022. *Nucleic Acids Research* 50(D1):D687–D692.
- Graves, A. 2016. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.
- Hao, M.; Gong, J.; Zeng, X.; Liu, C.; Guo, Y.; Cheng, X.; Wang, T.; Ma, J.; Zhang, X.; and Song, L. 2024. Large-scale foundation model on single-cell transcriptomics. *Nature Methods* 21:1481–1491.
- Ianevski, A.; Giri, A. K.; and Aittokallio, T. 2022. Fully-automated and ultra-fast cell-type identification using specific marker combinations from single-cell transcriptomic data. *Nature Communications* 13:1246.
- Islam, M. T., and Xing, L. 2023. Cartography of genomic interactions enables deep analysis of single-cell expression data. *Nature Communications* 14:679.
- Lan, Z.; Chen, M.; Goodman, S.; Gimpel, K.; Sharma, P.; and Soricut, R. 2020. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR)*.
- Liu, X.; Tao, Y.; Cai, Z.; Bao, P.; Ma, H.; Li, K.; Li, M.; Zhu, Y.; and Lu, Z. J. 2024. Pathformer: A biological pathway informed transformer for disease diagnosis and prognosis using multi-omics data. *Bioinformatics* 40(5):btac316.
- Luecken, M. D., and Theis, F. J. 2019. Current best practices in single-cell RNA-seq analysis: A tutorial. *Molecular Systems Biology* 15(6):e8746.
- Milia, M.; Tshimanga, L. F.; Mueller, H.; Atzori, M.; and Di Camillo, B. 2026. Integrating gene regulatory priors into transformer attention with scTransformer for interpretable scRNA-seq analysis. *arXiv preprint arXiv:2606.09558*.
- Raposo, D.; Ritter, S.; Richards, B.; Lillicrap, T.; Humphreys, P. C.; and Santoro, A. 2024. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*.
- Shazeer, N.; Mirhoseini, A.; Maziarz, K.; Davis, A.; Le, Q.; Hinton, G.; and Dean, J. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.
- The Cancer Genome Atlas Research Network; Weinstein, J. N.; Collisson, E. A.; Mills, G. B.; Shaw, K. R. M.; Ozenberger, B. A.; Ellrott, K.; Shmulevich, I.; Sander, C.; and Stuart, J. M. 2013. The cancer genome atlas pan-cancer analysis project. *Nature Genetics* 45:1113–1120.
- Theodoris, C. V.; Xiao, L.; Chopra, A.; Chaffin, M. D.; Al Sayed, Z. R.; Hill, M. C.; Mantineo, H.; Brydon, E. M.; Zeng, Z.; Liu, X. S.; and Ellinor, P. T. 2023. Transfer learning enables predictions in network biology. *Nature* 618:616–624.
- Tibshirani, R.; Hastie, T.; Narasimhan, B.; and Chu, G. 2002. Diagnosis of multiple cancer types by shrunken centroids of gene expression. *Proceedings of the National Academy of Sciences* 99(10):6567–6572.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Wang, T.; Shao, W.; Huang, Z.; Tang, H.; Zhang, J.; Ding, Z.; and Huang, K. 2021. MOGONET integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification. *Nature Communications* 12:3445.
- Yang, F.; Wang, W.; Wang, F.; Fang, Y.; Tang, D.; Huang, J.; Lu, H.; and Yao, J. 2022. scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data. *Nature Machine Intelligence* 4(10):852–866.
- Yang, R.; Dai, W.; Li, C.; Zou, J.; Wu, D.; and Xiong, H. 2023. scBiGNN: Bilevel graph representation learning for cell type classification from single-cell RNA sequencing data. *arXiv preprint arXiv:2312.10310*.